The Globalization Origins of Autocratic Rise: Engaged Reformers and the Post-Cold War Divergence

George Yean*

May 28, 2026

Abstract

Conventional wisdom holds that globalization promotes political liberalization and that inclusive institutions favor growth. Yet, unlike the previous period, the post-1990 era witnessed the consolidation and economic success of many autocracies, particularly in trade. I argue that globalization created structural conditions under which a specific regime type—economically engaged and reformed autocracies—gains a systematic trade advantage in competing for external demand, despite autocracies' weak domestic demand. Autocratic institutional features conventionally viewed as inferior in closed economies, such as centralized power or weak accountability, can become advantageous under globalization. Specifically, trade integration and domestic reform jointly supply credibility while autocratic institutions are less costly to form export competitiveness. This demand-competing advantage serves as a substitute for politically inclusive institutions. I connect macro-level patterns to micro-level mechanisms, showing how "autocratic engaged reformers"—accounting for over 90 percent of autocracies' economic output and extending beyond China and oil states—outperformed all other regime categories in exports. The findings offer a novel theory of autocratic rise rooted in the interaction between institutions and globalization, while revealing the limits of authoritarianism alone and the tensions globalization generates.

*George Yean is a PhD candidate at the Department of Government, Harvard University, gyean@fas.harvard.edu.

"...the view that democratic institutions are at best irrelevant and at worst a hindrance for economic growth has become increasingly popular in both academia and policy discourse ..."

– Acemoglu et al. (2019)

1 Introduction

Economic globalization markedly deepened after the Cold War (Baldwin 2016; Rodrik 2012), along with economic policy convergence (Simmons and Elkins 2004) and the expectation that economic liberalism advances political liberalization (Fukuyama 1989; Ikenberry 2001). The post-1990 era has instead produced widespread democratic backsliding, which is partially linked to globalization (Autor et al. 2020; Inglehart and Norris 2017; Rodrik 2017).[1] More striking is what this paper terms *autocratic rise*—many autocracies survived and reconsolidated, and statistically, have outpaced economic performance of democracies (Diamond 2015; Ekiert and Dasanaike 2024; Haggard and Kaufman 2016, Figure 1, also see more in Section 2).[2]

Why can autocracies rise economically alongside globalization? Studies that examine globalization and democratization (or domestic politics) (Autor et al. 2020; Milner and Mukherjee 2009; Rodrik 2012, Steinberg and Malhotra 2014) offer limited answers. The long debate on regime type and economic performance provides mixed evidence (Acemoglu et al. 2005; Barro 1996; Chandra and Rudra 2015; Olson 1993; Przeworski et al. 2000) or no comparison (Acemoglu et al. 2019); importantly, this literature emphasizes domestic factors and largely brackets the external environment such as globalization.

This paper places the contextual role of globalization at the center. I argue that autocratic rise is driven by a particular form of autocratic regime—those that are *both* economically engaged and reformed, or "engaged reformers." Globalization fulfilled this two-part *scope condition* under which autocratic engaged reformers gain an advantage over their democratic "counterfactual" in

[1]Scholars have called for "defining research programs" to understand this phenomenon (e.g., Drezner 2022).

[2]Autocracies' share may be under-estimated due to possible devaluation (Figure A.5), or possibly under-reported (Martinez 2021). Trade data reported by others is more accurate.

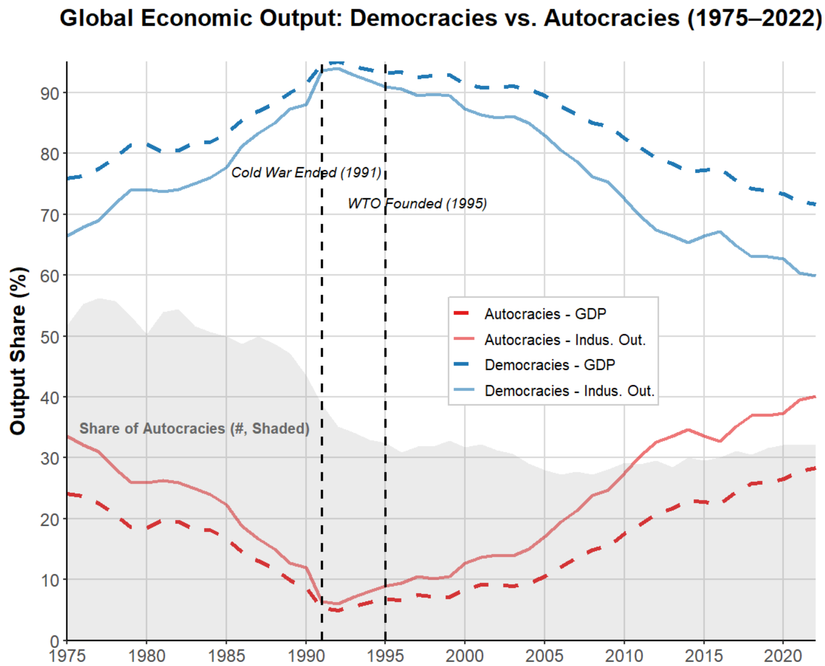


Figure 1: The Distribution of Production Between Democracies and Autocracies (World Bank). *Note*: I use a conservative measure for Autocracy (Freedom House (FH) Index ≥ 10), with similar patterns for using Polity or excluding China plotted in Appendix A.5. In 2020, China accounts for 62% of autocracies' GDP. Shaded area indicates the proportion of the number of autocracies.

external-demand competition, despite weak *internal* demand.[3] The pre-1990 Cold War era featured constrained economic exchange and regional blocs, when most developing countries—authoritarian or democratic—were closed or practiced inward-looking policies such as planned economy or import substitution industrialization (ISI), maintaining "trade skepticism" and eventually struggling with stagnation (Sachs and Warner 1995).[4] Under globalization, many soon embraced market-oriented and outward-looking reforms. Importantly, they were engaged by Western-led institutions such as the WTO and IMF, strengthening their credibility, reform progress, as well as access to markets, capital, technology, production networks, and policy ideas and information (Arias et al. 2018; Baccini and Urpelainen 2015; Baldwin 2016; Quinn and Toyoda 2007).[5] This is vital for autocracies because their non-inclusive political institutions undermine commitment credibility and domestic demand (Acemoglu and Robinson 2012, also see Figure 3).

The notion of "autocratic advantage" is not new (e.g., Wade 1990; Weede 1996). Recent work

[3]Demand stimulus may potentially extend to growth and innovation, e.g., China and Singapore.
[4]Except for some, e.g., export-oriented Asian countries and Chile, which closely align with my theory.
[5]For example, "Washington Consensus" prescribes outward-looking economic policies (PIIE 2021).

has documented, for example, their willingness to sign international agreements (Arias et al. 2018) and economic resilience (Lipscy 2018). Yet the mechanisms of the post-1990 autocratic rise remain underspecified. I argue that conventionally *inferior* autocratic institutions in closed economies can become export advantages in open-and-reform scenarios. This is because growth in inward-looking economies depends on domestic demand, autocratic institutional features—centralized authority, weak institutional constraint, or low accountability—are costly: they weaken commitment credibility and domestic consumption (Acemoglu and Robinson 2012). Globalization shifts the development logic outward, guides reforms, and provides ample room for states to alter trade patterns. Once engaged and reformed, now externally-oriented autocracies can supply sufficient credibility, while the same autocratic features—although still consumption-suppressing—remain less costly to form export competitiveness. This competitiveness is constructed through both *active* (e.g., industrial and export support, fiscal incentives, currency management) and *passive* channels (e.g., weak labor protection, low redistribution). For example, wage and welfare suppression only weakens domestic consumption, unless the resulting cost competitiveness captures more external markets. Conversely, hands-tying constraints that define democracies—deliberation, veto players—can blunt their capacity to compete for demand. The export advantage thus *substitutes* for inclusive institutions and compensates weak domestic demand, reconciling autocratic rise with conventional wisdom (Acemoglu et al. 2001; North and Weingast 1989) that inclusive institutions outperform.

Overall, globalization greatly enabled external-demand competition that advantaged reformed autocracies in export-led development. Consider several examples:[6] Asian "economic miracles," Qatar and the UAE, post-Soviet Azerbaijan and Russia, Ben Ali's Tunisia and dos Santos's Angola, and even, arguably, Modi's India. They differ sharply from traditionally repression-centric, isolated autocracies such as Mao, Saddam Hussein, or "Autocracy 1.0" (Lind 2025; Yang 2024). Instead, they emulated democracies' economic institutions and greatly owed their success to external demand.[7] By 2000, most post-communist states had reoriented trade to Western markets for growth (Åslund 2012), whereas countries like China and Vietnam heavily depended on exports (Feenstra and Wei

[6]Notable pre-1990 authoritarian successes include Pinochet's Chile or Franco's Spain. See Table A.1 for more examples.

[7]Before 1990, some inward-looking autocracies such as the USSR or Brazil did have short-lived growth success, but not markedly superior to democracies.

2010). Albeit resource-endowed, Qatar and the UAE took off only after the 1990s, leveraging WTO membership and heavy industrial investments (Hvidt 2013)—along with Angola to a lesser extent. Tunisia, under Ben Ali, prioritized export-led manufacturing and was the first in Africa to sign the Association Agreement with the EU, tripling exports before stagnation once it democratized.

I apply the theory to post-1990 globalization that profoundly transforms global trade, finance, migration, governance, or ideas.[8] Specifically, I focus on two trade-enhancing transformations that also jointly satisfy the scope condition above: trade integration (primarily WTO accession) and domestic economic reform. Because the theory centers on demand competition, I focus on exports that particularly drive growth, industrial development, and innovation (Bernard et al. 2018; Helpman and Krugman 1985).[9] I demonstrate that autocratic engaged reformers benefited the most from WTO accession and outperformed all other regime categories in exports—findings robust to various identification strategies and competing explanations (e.g., catching-up, commodity boom, importing credibility). Descriptive data shows that the same autocracies also outperform in mediating channels such as greenfield FDI, savings, and fixed investments. I further trace the mechanisms through three layers of evidence: sectoral patterns, a comparative case-study of China and India, and Japanese firms' investing preferences in Vietnam and the Philippines.

The theory therefore moves beyond conventional accounts of strong leaders, state capacity, resource-endowment, domestic reform (Allee and Scalera 2012), or simply China exceptionalism (although it well explains China's rise). For instance, former high-capacity socialist countries (e.g., North Korea and Uzbekistan) or resource-endowed states (e.g., Iran and Venezuela) that don't meet the scope condition underperform. By identifying how autocratic advantage is rooted in the interaction between domestic institutions and globalization, this paper provides an answer to the long-standing debate over regime type, institutions, and performance (Acemoglu et al. 2001; Fukuyama 1989; Ikenberry 2001; Rodrik 2012).[10] The theory also systematizes existing explanations for pre-1990 cases such as Korea, Chile, and Singapore (e.g., developmental state, see

[8] See Section 7 for more discussions.

[9] Long-run export growth (1992-2015) is strongly correlated with economic growth ($r = 0.74$, author's calculations). In Appendix, I also show that autocracies run higher external balances that are linked to global demand redistribution (Chinn and Ito 2021).

[10] For example, excessively high investments in China or Vietnam are unlikely without global market prospects.

Wade 1990), while explaining why many pre-1990 Latin American autocracies underperformed (they adopted inward-looking policies like ISI).

The argument carries implications for contemporary politics. Economic weakness erodes democratic foundations (Przeworski et al. 2000) and democracies' leverage abroad (Levitsky and Way 2006). From Russia's failed "shock therapy" to U.S. democratic deterioration, the delivery failure feeds authoritarian tendencies (Bruff 2014)—as citizens trade democratic accountability for performance in democratic backsliding (Graham and Svolik 2020; Krishnarajan 2023; Neundorf et al. 2024). Strengthened autocracies, in turn, reinforce a global authoritarian drift (Wright, Frantz, and Geddes 2013; Baturo and Tolstrup 2024), with implications for international security. This paper traces one source of these pressures to the unintended consequences of the liberal order: despite globalization's merits, it begets increased discontents, with "trade at the center (Stiglitz 2018)."

2 The Puzzle: Performance Divergence

I first document the patterns of the economic rise of autocracies in the post-1990 period. Figure 2 shows several economic measures more directly linked to globalization: the means of merchandise exports (% of GDP), trade balance (% of GDP), industrial output (% of GDP), and GDP growth rate of both democracy and autocracy groups. Since the early 1990s, the average performance of autocracies has diverged from that of democracies. These patterns are similar after removing developed countries, resources-oriented countries (such as Russia and the OPEC states), or China (see Figure A.4).

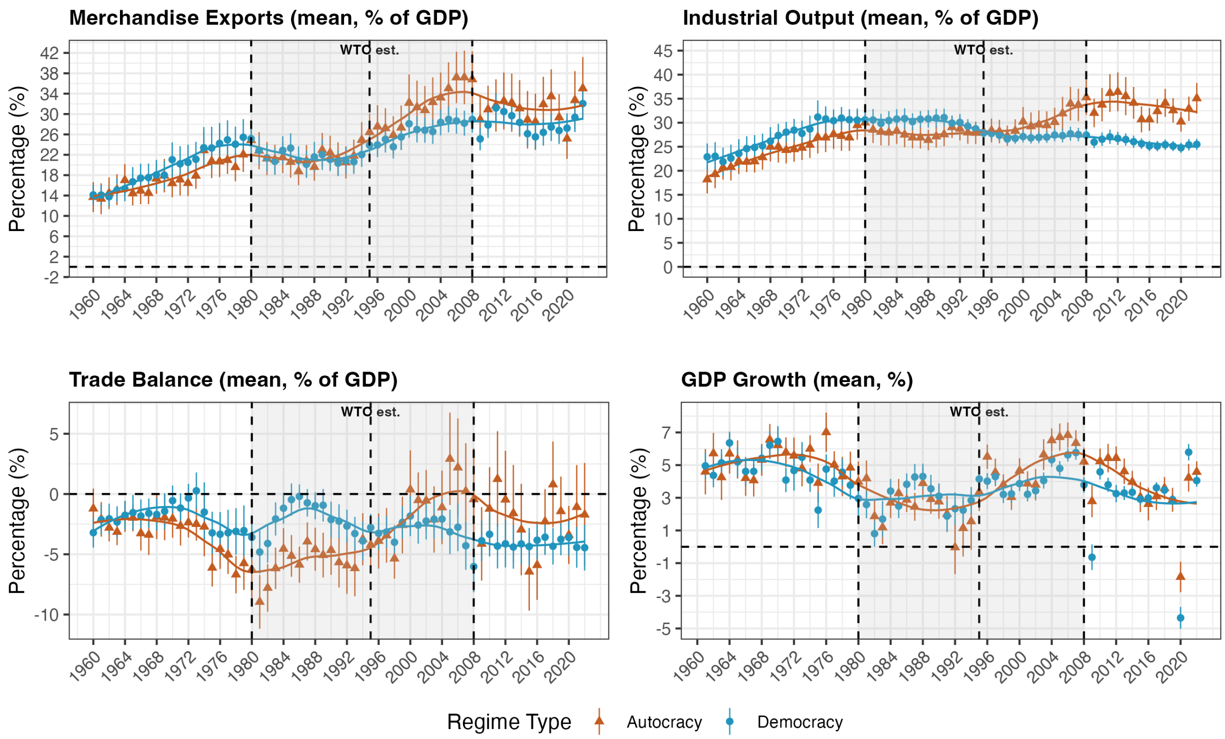


Figure 2: Mean Performance of Economic Indicators (World Bank) between Democracies and Autocracies (Polity <= 0). *Note*: Bars denote confidence intervals of means. Top and bottom 2.5% of observations are removed within each year. 1980 roughly marked the neoliberal turn. The patterns hold after removing developed states, or China, or Russia, or OPEC states (Figure A.4).

Figure 3 shows that autocracy is associated with stronger performance across nearly all major economic indicators between 1990 and 2020, except for consumption and taxation. A ten-unit decline in Polity (from 5 to -5), for example, is associated with approximately one percentage point higher GDP growth, eight percentage points higher industrial output, and nine percentage points higher exports.

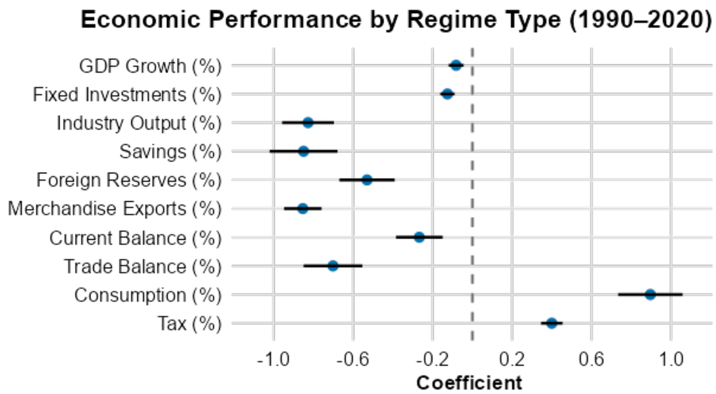


Figure 3: Regime Type and Major Economic Indicators (World Bank). *Note*: The bar plot means the percentage point change associated with one unit increase in Polity, controlling for GDP per capita and year fixed-effects for similar development-level comparisons. Correlations are robust to excluding China or OPEC states and including continent dummies and natural resource rents (%).

Connecting the Literature

The patterns above challenge conventional expectations about regime type and economic performance. Existing research generally finds weak or mixed effects of regime type on growth (Barro 1996; Przeworski et al. 2000). Democracies are expected to outperform because they protect property rights, foster political stability, invest in human capital, and support innovation (Baum and Lake 2003; Weingast 1995), while autocracies may generate growth by limiting redistribution and immediate consumption demands (Krueger 1974). Recent work finds that autocracy is associated with stronger growth only in the post-2000 period (Narita and Sudo 2021), but offers limited explanation. In trade, democracies are generally found to export more and adopt less protectionist policies, often due to stronger institutional environments (Atras 2015; Eichengreen and Leblang 2007; Yu 2010).

A related literature emphasizes the importance of inclusive institutions for long-run development (North 1990; Acemoglu et al. 2005), suggesting democracy's advantage. Another strand of literature identifies possible "autocratic advantages," including greater policy decisiveness and flexibility (Becker 2010; Easterly 2011), while recent work distinguishes reformed authoritarian regimes from purely repression-based ones (Lind 2025; Yang 2024).

Existing explanations, largely domestic in focus and relying on pre-2000 data, provide limited insights into the post-1990 autocratic rise. I contribute to this literature by incorporating globalization as an external context that reshapes institutional advantages. One puzzle, especially for institutional theories, is that even in the post-1990 period, democracies on average still exhibit higher institutional quality; nevertheless, they underperform.

3 Age of Globalization: "Autocratic Advantage" Revisited

The intuition that some authoritarian regimes may enjoy developmental advantages is not new. Nobel laureates Michael Spence and Gary Becker, for example, have observed that rapid growth requires "strong political leadership (Spence et al. 2008)" or "not heeding constraints in promoting agenda (Becker 2010)." Historical cases such as South Korea and Chile suggest that strong performance under authoritarian rule depends on three general conditions: credible economic institutions, external market access, and state-led trade-enhancing policies.[11] This section theorizes when that intuition should hold.

The central claim is that, under post-1990 globalization, a subset of autocracies—once they are economically engaged and reformed—obtain distinct advantages in competing for external demand. I refer to these regimes as "autocratic engaged reformers." Autocracy is not inherently superior. In closed or unreformed economies, non-inclusive authoritarian institutions often generate familiar pathologies, such as weak credibility, extraction and misallocation, and weak domestic demand. But once engaged and reformed, some authoritarian features can translate into export advantages.

Export receives wide scholarly attention because demand-side stimulus promotes growth by inducing investment and supporting scale economies (Harrison and Rodriguez-Clare 2010; Keynes 1936; Porter 1998), whether in the case of export-oriented successes or wartime economies. This is critical for autocratic regimes inherently weak in domestic demand.[12]

3.1 New Economic Context: the Globalized Economy

I first examine globalization as a new economic context. Compared to the pre-1990 Cold War era, the globalized economy witnessed unprecedented expansion in trade and capital liberalization and

[11] These three conditions align well with the existing literature: institutionalism (North 1990), growth theories (Solow 1957; Romer 1986), and development theories that underscore the state role (Evans 1995; Haggard 1990).

[12] For example, even a slowdown in exports causes social instability in China (Campante et al. 2023).

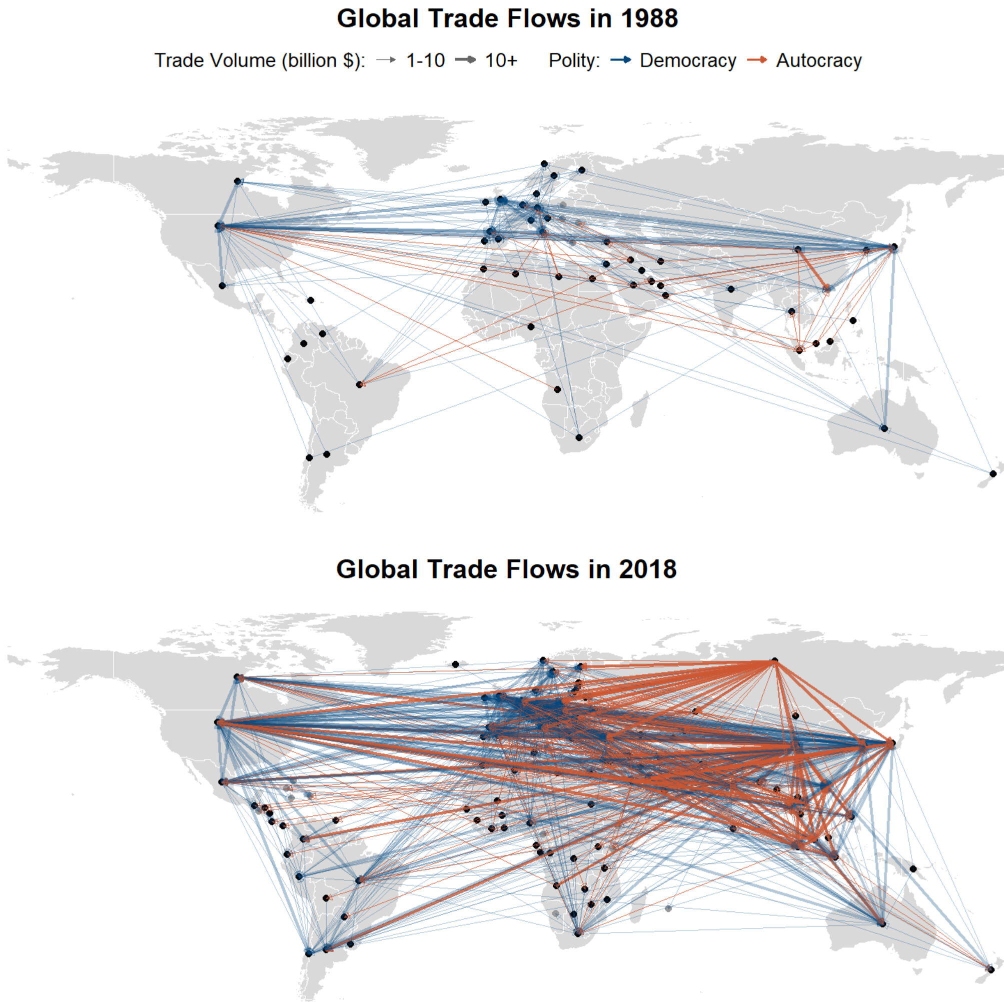


Figure 4: Global Bilateral Trade Flows (1988 vs. 2018). *Note*: For visualization and export-emphasizing purposes, the plot displays only the larger of bi-directional flows colored in exporter's Polity. For example, red lines denote export flows from autocracies (FH ≥ 10). Before 1990, exports are mostly dominated by democracies.

globalized production, often called "hyperglobalization" (Rodrik 2011, also see Figure 4). This, I argue, resulted in two structural shifts.

First, globalization fundamentally reoriented the logic of development outward. Relative to the pre-1990 world of limited exchange, regional blocs, and prevalent inward-looking development strategies (e.g., planned economy, ISI), the post-Cold War period is characterized by deep trade integration, capital mobility, globalized production, and rapid policy diffusion (Bernard et al. 2018; Simmons and Elkins 2004). Growth depends less on mobilizing domestic demand than on securing access to external markets, capital, technology, and production networks (Baldwin 2016; Bernard et al. 2018). The rapid transfer of productive know-how enables the swift ascent along the value chains (Baldwin 2016), while expanded market access stimulates firm expansion and realizes scale advantage (Krugman 1979; Melitz 2003), which drive output and innovation (Atkeson and Burstein 2010; Burstein and Melitz 2013; Grossman and Helpman 1991). Another key factor is information. International institutions widely guided governments how to reform and design policies. For example, the "Washington Consensus" prescribes outward-looking economic policies (PIIE 2021), while the World Bank recommends a combination of institutional building and policy support to boost growth.[13] Finally, globalization subjects governments to external stakeholders—including international investors and institutions—that leaders can not ignore as they once could in domestic economies (Arias et al. 2018; Jahn 2006).

Second, a globalized economy creates ample policy room for states to enhance export competitiveness. While conventionally regarded as mutually beneficial, free trade models assume idealized conditions such as perfect competition and few frictions, which often do not hold in practice (Stiglitz 2018)—partly explaining why moving from autarky to open trade is a one-time gain, with the long-term effect being debatable (Garrett 2000). Institutional differences, macro-financial conditions, and policies like currency devaluation, subsidies, and wage suppression can all create artificial advantages especially when technology converges (Antràs 2015; Brander and Spencer 1985; Corsetti et al. 2007).[14] This is especially pronounced when a key free-trade assumption—balanced trade (or exogenous imbalance) on which comparative advantage materializes—rarely holds.[15] Eaton and Kortum (2002) similarly describe a country's export competitiveness as "technology adjusted for costs," echoed by others (Bernard et al. 2003; Melitz 2003).[16] Epifani and Gancia (2017)

[13] "Global Economic Prospect" Report, World Bank, 2024.

[14] For example, in the classical Eaton-Kortum model (2002) which assumes a country takes a random productivity draw for goods from a Fréchet distribution which generates a country's comparative advantages. Currency devaluation ends up acquiring more competitive advantages. Moreover, Arkolakis et al. (2018) show that MNCs choose production location l based on final unit cost: $C_{il} = \frac{\gamma_{il} w_l \tau_l}{z_l}$, where γ_{il} is the foreign production cost, w_l is local wage, τ_l is trade cost, and z_l is firm productivity which can be from within-network technology transfer.

[15] E.g., see the discussion of global imbalances in Blanchard and Milesi-Ferretti (2009) and Obstfeld and Rogoff (2009).

[16] In Eaton-Kortum model (2002), the proportion of country n's total expenditure imported from country i is: $\pi_{ni} = \frac{T_i (w_i d_{ni})^{-\theta}}{\sum T_h (w_h d_{nh})^{-\theta}}$, where T_i represents technology and w_i represents wage or factor cost. Given technology convergence (e.g., due to the diffusion by GVC), wage determines production location. The Melitz model (2003) similarly specifies that firm's profit ($\pi(\phi) = [(\frac{\phi}{\phi^*})^{\sigma-1} - 1] w f$), which determines firm's entry into export market, is determined by wage w. Bernard et al. (2003) show that decreased wage increases competitiveness and the range of exports.

further demonstrate that currency undervaluation leads to trade surpluses and production agglomeration. Cost advantage, additionally, can be enhanced by one's infrastructure, regulations, resource endowment, and even work culture.

In sum, a globalized economy shifts development logic outward, while providing greater information and room for states to effectively learn and practice trade-competitiveness-enhancing policies. Rather than relying on domestic demand and imposing domestic distributional costs (e.g., subsidizing producers), the focus extends to competing for external markets, technology, and investment, and externalizing costs onto foreign states.

3.2 Autocratic Advantages in Open-and-reform Conditions

How, then, do autocratic engaged reformers acquire export competitiveness in a globalized economy? Before the 1980s, most autocracies were either unreformed or inward-looking (Sachs and Warner 1995) and eventually failed (e.g., China's Mao, Egypt's Nasser, Ghana's Nkrumah). Subsequent learning and reforms (e.g., improving rule of law and protecting property rights (PR)) created autocratic engaged reformers: a regime form that can participate credibly in global markets while preserving less constrained political institutions.

Scope condition. I argue that autocratic export advantages are conditional on engagement and reform. Because exporters must be able to access outside markets and view the economic environment as sufficiently credible, autocracies that preserve state monopolies, insecure property rights, or trade isolation can hardly benefit from globalization—they suppress domestic demand or simply produce predation. Under globalization, conventionally inferior autocratic characteristics generate lower costs for reformed autocracies than comparable democracies, for example, to weaken labor-rights protection or channel resources toward export sectors, at the cost of domestic consumption.[17] Additionally, since Reagan and Thatcher's neoliberal turn, (semi)authoritarian measures are often less costly for implementing reform and managing resistance (Bruff 2014).

However, the engaged-reformer scope condition, as a minimum threshold, does not automatically reflect export competitiveness, which requires more than basic institutional arrangements.

[17] Note that this advantage is not pure state-led industrialization, nor does it necessarily require active intervention; it can be passive, for example, through weak labor protection.

For example, many countries with meaningful rule-of-law and WTO membership nevertheless underperform. Selecting into WTO accession and domestic reform reflects a multitude of historical, domestic, and geopolitical reasons rather than pre-existing export competitiveness (see Section 4), especially when comparing within engaged reformers across regime types. Below, I theorize four channels explaining how regime characteristic1234s amplify export competitiveness.

Centralized power. Autocratic states concentrate coordination capacity, as opposed to more fragmented democratic systems (Diamond 2015). Centralized power in a closed or unreformed state often leads to expropriation and resource misallocation. However, in a globalized economy, centralization enhances reform-minded leaders' position to push for reforms, deploy concerted policies, and respond swiftly to changing global markets, without delays from fragmented branches (Hall and Soskice 2001; Kohli 2004). Autocracy's relative leadership longevity can facilitate consistent and long-term economic planning (Wade 1990), creating a predictable business environment (Haggard 1990). Moreover, external shocks can be resisted more effectively than in more "hands-off" democracies (Shih 2020). Apart from the authoritarian examples above, even democratic but more centralized postwar Japan and Modi's regime were able to implement impressive trade-boosting policies.

Weak institutional constraints (*ex ante*). Autocracies operate with weak institutional constraints from constitutions, legislatures, opposition parties, and norms (Levitsky and Way 2010). Such governments can cause unchecked predation in closed economies, but in open-and-reform scenarios, they can more readily divert resources from welfare or other interest-balancing programs to export-oriented projects with little obstruction. Autocracies' larger win-set (Putnam 1995) can better appease stakeholders and sign favorable international agreements. Empirically, autocratic states are found to establish more special economic zones offering tax breaks and looser regulations (Allen and Ge, working paper) and are more flexible in controlling exchange rates (Steinberg and Malhotra 2014) and financial institutions (Brune et al. 2001). In contrast, democracies experience more financial instability, primarily due to their weaker controlling abilities (Lipscy 2018).

Limited accountability (*ex post*). Even reformed autocracies remain more insulated from voters and corporatist, labor, or environmental groups (Krueger 1974; Rodrik 1999). While non-inclusiveness hampers domestic consumption (Chandra and Rudra 2015), when outward-oriented, it allows them to pursue a broader range of market-favoring policies for external markets, even when socially unpopular or costly (Quinn and Woolley 2001). Promotion is more based on performance in authoritarian regimes (Jensens and Malesky 2018) which can more easily impose savings for investments. With weak labor protection, autocracies can enhance policy flexibility and cost competitiveness. In contrast, democracies are often pressured to meet immediate demands for consumption or redistribution—particularly in poorer countries (Zakaria 1997)—which can undermine market efficiency (Huntington 1968; Sah 1991). For example, many Latin American democracies have stricter labor regulations than OECD countries (Feierherd 2024).

Mercantilist mentality. Ideas shape practices. Authoritarian leaders embraced reforms because of legitimacy crises, international pressures, and economic incentives (Baturo and Tolstrup 2024; Levitsky and Way 2006; Haggard and Kaufman 2016), which can result in a more mercantilist manner. Democracies, by contrast, tend to be more economically liberal (Milner and Kubota 2005). While conventionally unrecommended, mercantilist policies such as directed credit and selective regulatory enforcement can foster domestic industries or incentivize firms to produce locally, as exemplified in China's automobile industry and India's electronics sector under Modi.

The list can extend to resource endowment that creates export foundations for some autocracies,[18] as well as consumption-disregarding culture and norms, particularly in former socialist economies (Fitzpatrick 1999; Nove 1986). The implication is not merely that some autocracies export more. It is that post-1990 globalization created conditions that altered the political economy of regime performance. That shift leads to the post-Cold War association between autocracy, trade, and broader economic strength.

3.3 (to Appendix) A Model of Engaged Reformer and Trade Facilitation

To formalize the theoretical logic, I present a two-player model in which regimes acquire their levels of trade-facilitating policies (or trade manipulation vis-à-vis free trade practices), proactive or passive (e.g., industrial policy, wage suppression, or currency devaluation) to maintain political

[18]Resource abundance can be both "resource curse" in a closed, unreformed economy (Ross 2001) and enabling strategic reinvestment of rents, as exemplified by the active role of investing bureaus in Qatar and the UAE.

support from selectorate (Bueno de Mesquita et al. 2003).[19] The model combines leader perception, institutional structure, and the cost and benefit of manipulation, illustrating that autocratic engaged reformers are more likely *and* easier to obtain a higher level of trade facilitation.

Setup. Consider otherwise same countries – except for regime type autocracy $R(\phi = 0)$ or democracy $R(\phi = 1)$ – choosing a level of trade manipulation $T \in [0, 1]$. I assume that unless regime is both trade engaged ($w = 1$) and reformed ($r = 1$), trade manipulation is barely effective. I assume that trade manipulation increases export performance and yields private returns $\alpha T(w = 1, r = 1)$ to the regime, but it incurs institutional and international reputation costs as follows:

$$c(\phi, T) = \rho(\phi)T + \gamma T \tag{1}$$

where institutional cost coefficient is denoted by $\rho(\phi) = \rho_0 + \rho_1\phi$ with $\rho_1 > 0$, while $\gamma > 0$ denotes positive reputation cost coefficient. Institutional costs are higher in democracies due to institutional constraints, veto players, and transparency. I assume reputation costs are the same, although democracy may perceive more.

Regime utility depends on whether it secures domestic support ($S = 1$) or not ($S = 0$). If support is secured, regime gains a normalized utility 1, and 0 otherwise. Regime's total utility is given by:

$$U_R(\phi, T) = \begin{cases} 1 + (\alpha - \rho(\phi) - \gamma)T & \text{if } S = 1 \\ 0 & \text{if } S = 0 \end{cases} \tag{2}$$

Selectorate receives baseline economic benefits $d(\phi)$ from the regime alone (e.g., non-export revenue-based public services) and additional trade-related gains bT. Selectorate generates costs $C^*(\phi)$ by supporting the regime. I assume $C^*(0) > C^*(1)$ because autocracies require stronger obedience absent electoral legitimacy. Also assumed is $d(0) \leq d(1)$ because democracies tend to deliver stronger baseline domestic performance due to more inclusive political institutions (Ace-

[19] Passive manipulation like disallowing labor unions also generates costs.

moglu et al. 2001). Additionally, autocratic leaders may perceive more severe consequences from losing power, as well as lower baseline benefits, which increases their perceived performance threshold required, that is, $P^*(0) > P^*(1)$.

Thus in the eyes of leaders, selectorate supports the regime if and only if total received benefits exceed the perceived regime-specific support threshold $P^*(\phi)$. This condition reflects a minimal level of economic performance threshold leaders need to keep power.

$$S = 1 \quad \textit{if and only if} \quad d(\phi) + bT \geq P^*(\phi). \tag{3}$$

Optimization. Anticipating selectorate's response, regime maximizes utility by solving:

$$\max_{T \in [0,1]} U_R(T, \phi) = \mathbb{I}[rw = 1] \cdot (1 + (\alpha - \rho(\phi) - \gamma)T) \quad \text{s.t.} \quad T \geq \frac{P^*(\phi) - d(\phi)}{b}. \tag{4}$$

If $rw = 0$, regime does not manipulate trade. The regime chooses the minimal level of T that satisfies the selectorate's support constraint. Let $T^*(\phi)$ denote the regime's equilibrium manipulation level. Since democracy faces higher institutional and reputational costs and offers leaders fewer private trade benefits, there is less incentive for them to engage in trade manipulation. Drawing on empirical evidence, it's reasonable to assume $\frac{\partial U_R}{\partial T} = (\alpha - \rho(1) - \gamma) \leq 0$. In contrast, autocratic leaders likely face lower institutional and reputational costs than trade benefits: $\frac{\partial U_R}{\partial T} = (\alpha - \rho(0) - \gamma) \geq 0$. Based on the linear function of $U_R(T)$, democracy chooses the minimum value $T^*(1) = \frac{P^*(1) - d(1)}{b}$ to ensure support, while autocracy chooses a value larger than threshold $T^*(0) \geq \frac{P^*(0) - d(0)}{b}$.

Proposition 1. Autocracy chooses a higher level of trade manipulation than democracy.

$$T^*(0) \geq \frac{P^*(0) - d(0)}{b} \geq \frac{P^*(1) - d(1)}{b} = T^*(1) \tag{5}$$

Proof. Since $P^*(0) > P^*(1)$ (higher perceived support threshold in autocracy) and $d(0) \leq d(1)$ (lower baseline domestic performance in autocracy), it follows that $P^*(0) - d(0) \geq P^*(1) - d(1)$. Dividing both sides by $b > 0$ leads to $T^*(0) \geq T^*(1)$. Thus, autocracy chooses a higher level of trade manipulation.

Proposition 2. As institutional constraints in autocracies increase (e.g., due to continued reforms), trade manipulation decreases.

Proof. If $\rho(0)$ increases, $(\alpha - \rho(0) - \gamma)$ decreases — meaning the slope of utility becomes less positive or even negative. Thus, increasing $\rho\phi$ makes trade manipulation less attractive, so less likely to choose a high $T^*(0)$. That is:

$$\frac{\partial T^*(0)}{\partial \rho(0)} < 0. \tag{6}$$

Proposition 3. If a stronger global detection of trade violation increases reputational costs (γ), trade manipulation decreases. For autocracy, total gains below zero threatens regime survival.

Proof. Reputational cost enters linearly in the slope $(\alpha - \rho(\phi) - \gamma)$. An increase in γ decreases the marginal return to manipulation for both democracies and autocracies. As γ increases, regimes may reduce manipulation to meet only the minimum support threshold. Particularly for autocracies, if $(\alpha - \rho(0) - \gamma)$ is already small, a rise in γ can shift the slope negative, making even the minimum required $T^*(0)$ insufficient to maintain selectorate support, thus threatening regime survival.

This stylized model formalizes the intuition that autocrats, facing greater perceived vulnerability and fewer institutional constraints, engage in higher levels of trade distortion. It explains why reformed autocracies may manipulate trade more aggressively – doing so not despite but because of their heightened need and lower costs to deliver tangible performance gains in lieu of democratic legitimacy.

4 Autocratic Engaged Reformers in the Post-1990 Globalization

This section applies the theory to post-1990 globalization, which realized the scope condition of autocratic engaged reformers. Among the numerous changes that occurred during post-1990 globalization, two key trade-enhancing transformations are trade integration and domestic reform, which closely align with the development literature and the scope condition. Other changes are either less relevant (e.g., migration and culture) or predicated on trade integration (e.g., export-oriented FDI). More is discussed in Section 7.

Trade Integration

Since the early 1990s, the global trade regime confined to the Western Hemisphere had begun to expand. Trade integration served as the foundation for participating in globalized production, foreign investments, and technological diffusion, and was embodied in trade agreement proliferation (see Figure 5).[20] Of all, the WTO plays a major role in global trade integration (Bagwell and Staiger 2002), praised as the "most heralded commercial agreement in history" (Goldstein et al. 2007).[21] The WTO, or its predecessor GATT, stipulates the prohibition of discriminatory tariffs among members, facilitating market access with progressively lower tariffs.

[20] What distinguishes post-1990 global trade from previously also includes the spread of the global value chain (Baldwin 2016, also discussed in the "New New Trade Theory").

[21] Regional trade deals usually build on top of WTO principles to address specific trade issues: e.g., sector-specific trade and dispute resolution.

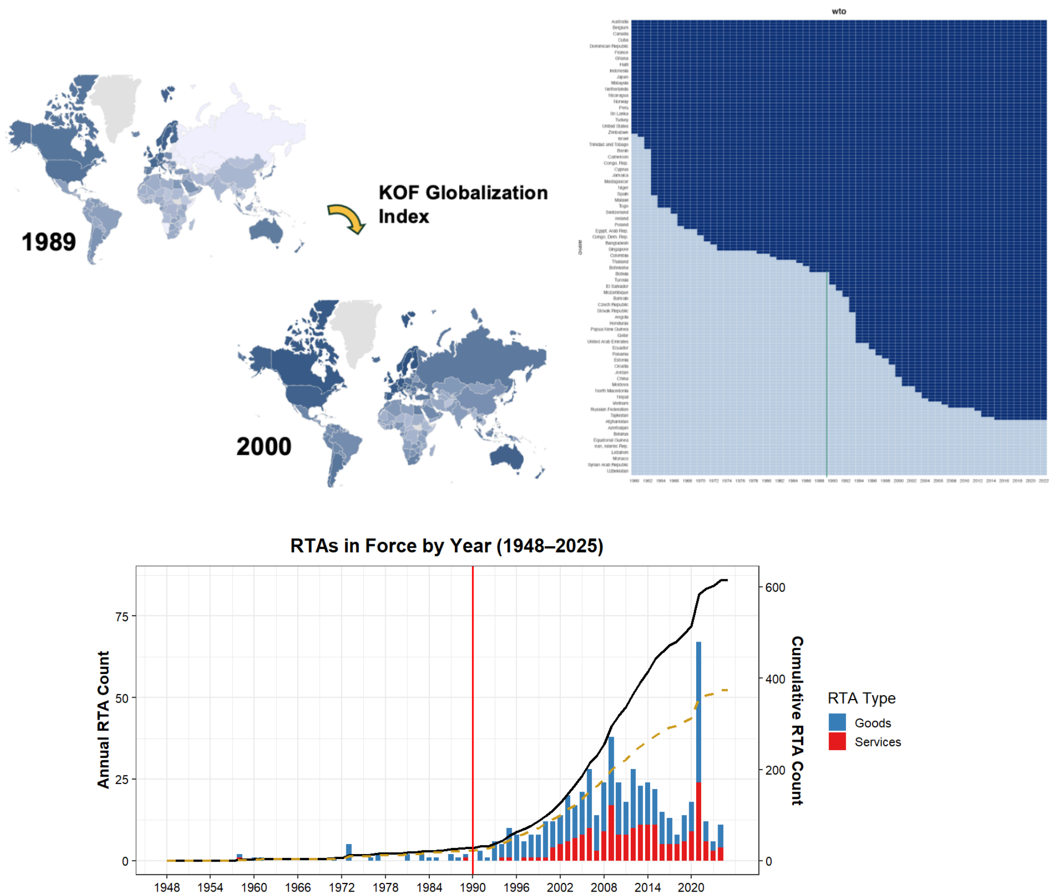


Figure 5: Post-Cold War Economic Globalization Expansion. *Note*: The figures respectively depict: KOF Globalization Index; GATT/WTO Expansion; RTA Proliferation.

Contrasting a quite conservative pre-1990 accession process, the WTO integrated many autocratic and newly democratized countries after 1990.[22] The number of members almost doubled, increasing from 88 in 1985 to 164 in 2020. This allowed autocracies greater access to global markets and materialized their advantages through trade unseen during the Cold War. For example, by 2000, most post-communist countries depended their growth on exports to Western markets, especially for the most reformist (Åslund 2012). Of note, although some autocracies joined late or still haven't joined, many enjoyed "semi-engagement": for example, the Most Favored Nation (MFN) status—a WTO principle—from major western countries,[23] PTAs and RTAs, a globalized commodity market, and the spillover from joiners—all almost impossible before 1990.

[22]The 1950s-joiners were mostly advanced democracies such as Germany, Japan, Italy, Austria and Sweden, while the 1960s-joiners were mostly formerly colonies by "automatic accession" under Article 26:5(c). The 1970/80s saw much stagnant accession progress.

[23]See https://1997-2001.state.gov/regions/eap/fs-mfn_treatment_970617.html.

Studies have found that the WTO substantially increases trade (Goldstein et al. 2007). Davis and Wilf (2017) simulate that China and Mexico's export booms would have occurred earlier had they joined earlier.[24] Apart from market access, the WTO also strengthens institutional credibility, particularly for politically dissimilar countries (Carnegie 2014), implying that autocracies may benefit more from joining a democracy-dominated club. Such international institutions enable states to make credible commitments (Hafner-Burton and Schneider 2019), which attract FDI along with technology (Buthe and Milner 2008). This is particularly important in an era when multinational investors' confidence shapes trade (Bernard et al. 2018). The WTO also offers programs such as the Trade Policy Review Mechanism (TPRM) for policy learning and diffusion.

Institutional Reform

Stable autocrats have long understood how bad excessive exaction is for survival (Olson 1993). Starting from the 1980s, under multifaceted pressure ranging from economic to ideological, many autocracies (as well as democracies) in the developing world have followed the "Washington Consensus," initiating various degrees of market-oriented reforms (Quinn and Toyoda 2007). These reforms include establishing the rule of law and privatizing state-owned enterprises, as well as adopting business-friendly policies, for example, PR protection and financial and labor market deregulation. Some extended liberalization by opening up trade and capital accounts (Milner and Mukherjee 2009). Liberalization reform under globalization is outward-looking: many developing countries moved from planned economies or ISI to more export-emphasizing models, including those already in the GATT (e.g., Egypt, Morocco) and later joiners.[25] Moreover, WTO accession's conditionality and the "imported credibility" reinforce domestic institutions (Arias et al. 2018).

Figure 7 shows the historical trends of two major institutional measures: PR protection and rule of law. The former focuses on the protection of investments from expropriation, while the later emphasizes contract enforcement and dispute settlement (Pandya 2016). These institutions foster growth (North and Weingast 1989) by stimulating domestic entrepreneurship and multinational

[24] Per data, even resource-oriented countries such as the UAE and Oman experienced an immediate export boost upon WTO accession after years of weak exports.

[25] For example, states choose to join the WTO hoping to increase exports (Pelc 2011). Even developed countries like Canada maintain agencies like Export Development Canada.

firms to set up productive facilities (Atras 2015), greatly boosting exports.

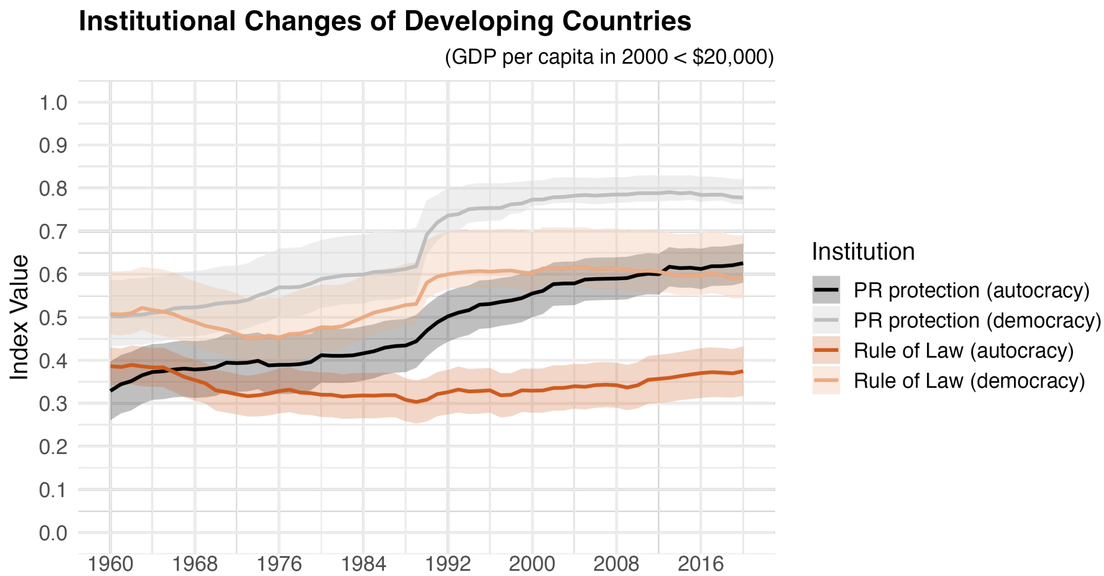


Figure 6: Average Rule of Law and Property Rights Protection. *Note*: Autocracies/democracies are roughly divided by Polity in 1991 (when the USSR dissolved) to ensure temporal country data integrity. Shadows denote 95% CIs. Cross-country variations exist within each category.

If autocracies remain unreformed, exposure to global trade will not help much. Turkmenistan and Azerbaijan are comparable cases: similar in rich natural resources,[26] Polity score, geographical location, culture, race, population, and per capita income in the 1990s. Yet, Turkmenistan has significantly lower PR protection than Azerbaijan (0.16 vs. 0.66). While neither has joined the WTO, both were semi-engaged through MFNs and regional trade. From 1992 to the mid 2010s, their exports grew 11 and 24 times, respectively.

However, unlike democracies, which usually follow more hands-off reforms reliant on markets (Harvey 2005), autocracies usually do so selectively and gradually, often with less political reform. China, for instance, implemented rule of law selectively only for foreign investments and regime durability (Wang 2015). While allowing trade flows, many autocracies have strictly controlled exchange rates and capital accounts, shielding themselves from external shocks (Kuzio 2020; Steinberg and Malhotra 2014) and facilitating mercantilist policies. Additionally, autocracies

[26] Turkmenistan is slightly better: 3.8% of world's natural gas reserve and 0.04% in oil, while Azerbaijan has 0.5% of world's reserve in natural gas but higher (0.42%) in oil (U.S. EIA).

often control strategic and politically sensitive sectors and actively support industrial development, effectively practicing "embedded liberalism" that fuses markets with state goals.

Combining Trade Integration and Reform

The theory requires both trade integration and reform. Then autocratic engaged reformers can "embed" authoritarian characteristics into economic liberalization to enhance export competitiveness. Non-reformed WTO members are not conducive to substantial trade growth (Allee and Scalera 2012; Tang and Wei 2009), nor are trade-isolated reformers. This differs from the literature that primarily emphasizes domestic reforms (e.g., China's adaptive institutions, Ang 2016).

	In WTO (before 2015)	**Not In WTO**
Moderate-high Institutional-levels	**"Engaged Reformers"** Angola (15.2), Bahrain (5.4◊), Cambodia (21.2), Cameroon (2.6), Chad (22.6▲), China (22.9), Congo Rep. (8.6), Djibouti (10.2), Egypt (8.1), Jordan (6.2), Kazakhstan (21.5), Kuwait (9.7◊), Lao (12.7), Mauritania (5.1), Morocco (6.1), Oman (8.5), Qatar (33.6◊), Russia (8.8), Rwanda (12.2), Saudi Arabia (6.8◊), Singapore (5.1◊), Tanzania (10.1▲), Thailand (5.8), Togo (5.2), United Arab Emirates (13.3◊), Uganda (9.5▲), Vietnam (46.2▲) (*Former Aut.*: Brazil, Chile, Hong Kong, Indonesia, Malaysia, Mexico, South Korea, Spain, Tunisia ...)	**"Unengaged Reformers"** Afghanistan (2.9▲), Algeria (5.2), Azerbaijan (23.8▲), Belarus (12.5), Equatorial Guinea (201.2▲), Ethiopia (12.4▲), Iran (4.5), Iraq (8)
Low Institutional-levels	**"Engaged Non-reformers"** Congo Dem. Rep. (6.3▲), Myanmar (17.8▲), Swaziland (2.7), Tajikistan (3.2), Venezuela (4.5)	**"Unengaged Non-reformers"** Cuba (3.4), Eritrea (6.9), Libya (2.8◊), North Korea (4), South Sudan (NA), Sudan (9.8), Syria (0.4), Turkmenistan (10.9), Uzbekistan (4.1), Yemen (4.6)

▲*: Average GDP per capita under $200 in 1991-1993,* ◊*: above $5,000*; Numbers in parentheses are export increase from the early 1990s to mid 2010s.

Table 1: Typology of Autocracies. *Note*: autocracies are roughly defined as those with average Polity ≤ 0 in 2000-20. Poor institution refers to institutional levels that are below the thresholds for VDem PR protection and rule of law (see Appendix B.3). Together, "engaged reformers" accounted for over 97% of autocracies' GDP in 2015.

Table 1 roughly classifies all existing autocracies into a 2x2 table by institutional levels and WTO membership. Many fall into the category of "engaged reformers," meaning they have achieved meaningful levels of institutions and are in the WTO.[27] As a face validation, many in this category, beyond oil states or China, seem to perform well.[28] In contrast, most states in the other three categories underperformed. The same resource-rich autocracies—Algeria, Iran, Iraq, and Venezuela—performed worse than engaged reformers such as Qatar, Kuwait, and Morocco. Many democratic engaged reformers underperformed, including Argentina, Brazil, Colombia, Kenya, Mexico, Nigeria, Pakistan, Peru, the Philippines, South Africa, and Ukraine. 25 out of 40 fastest growing countries (1992-2015) are autocracies, which make up only 25% of total countries.[29]

Autocracies may particularly require trade integration because their political non-inclusiveness is often weak in fostering domestic demand. Joining international institutions also adds more credibility to autocracies (Arias et al. 2018). Autocracies still possess lower average institutional levels (Figure 7), which should predict lower performance in closed economies. This implies that trade integration is more indispensable for their economic rise.

In sum, I expect that reformed autocracies that possess more discretionary power and fewer constraints should gain the most in trade in the post-1990 period (i.e., experience the largest effect of WTO accession on exports). Yet this same logic also implies that institutions, at sufficiently high levels, progressively constrain leaders' discretion by dispersing authority across more procedural rules (e.g., institutionalized party consultation or administrative transparency). Autocratic export advantages should therefore be strongest under intermediate institutionalization: meaningful reform to supply credibility, but not so much constraint on leaders—that is, not too low (insufficient credibility) and not too high (excessive constraint). Accordingly, I define advantage cases (engaged reformer) as in-WTO regimes with moderate institutional levels.[30] Finally, I expect that autocratic engaged reformers should outperform comparable democracies in absolute export levels.

In Appendix B.1, I present an extended trade model based on the classic Eaton-Kortum (E-K) model (2002) to illustrate the predictions of the above hypotheses. The E-K model captures the

[27] Based mainly on institutional levels at the bottom 20 percentile among developing countries in 2010. See Appendix for more details.

[28] Some countries, such as Cameroon, Mauritania and Togo, do not stand out (although they may still perform better than their "democratic counterfactual"). These newly independent colonized countries automatically joined the GATT; many have ostensibly "copied" economic institutions from former colonizers, but with few substantive reforms (Allee and Scalera 2012).

[29] Based on the WDI data.

[30] Below, I show the results are not sensitive to simple low/non-low classifications.

determinants of bilateral trade flows such as technology, production cost, trade cost, and comparative advantage, making it particularly suitable for my case. The model is enhanced with three additional variables: institution, trade engagement, and autocratic advantages, with proper functional forms.

5 Empirics: Testing "Engaged Reformers"

Although the main analysis focuses on trade outcomes, I first examine whether autocratic engaged reformers exhibit economic patterns consistent with the mechanisms described in autocratic advantages.

Regime Category	Growth %	Indus. %	Consump. %	Greenf. FDI %	Fix. Inv. %	Sav. %	Exp. %
Aut. Engaged Reformer	**4.6**	**35.8**	58.4	**6.0**	**25.6**	**25.3**	**32.8**
Dem. Engaged Reformer	4	25.4	**69.7**	5.3	22.2	15.3	25.3

Table 2: Major Performance of Engaged Reformers (Autocratic and Democratic, 2000-20). *Note*: Averaged over in-WTO country-years among 133 developing countries (< $20,000 average income in 2000). Institutional cutoffs see below. Source: World Bank.

Descriptive comparisons (Table 2) confirm that, among developing WTO members during 2000-20, autocratic engaged reformers display more export-oriented economic profiles (including mediating channels such as investment and savings that connect regime characteristics to exports) than comparable democracies.

5.1 WTO Accession

I next examine the effect of joining the WTO—the most significant embodiment of trade integration. I empirically choose the cutoff year 1990 because it not only witnessed a dramatic global political shift—the end of the Cold War—but also an economic shift often termed as "hyper-globalization": the unprecedented proliferation of trade agreements and flows of goods and capital (Pandya 2016). Additionally, there had been worldwide domestic reforms and democratization. The choice is also for empirical convenience. In Figure 7, I also conduct the time-rolling estimates.

New WTO joiners

Between 1990 and 2020, in total 64 countries (with over half a million trading-pairs) joined the WTO/GATT, and almost all were developing countries in 1990 (except Liechtenstein). Of them, 25 (Freedom House Index ≥ 8) or 18 (Polity ≤ 0) were autocratic states in 1992.[31] These countries do not account for the majority of existing autocracies across the world, but include major autocracies such as China, Russia, Saudi Arabia, Vietnam, United Arab Emirates, Qatar, Oman, Kazakhstan, Tajikistan, Kyrgyz, Bahrain, Tunisia, Angola, Lao, Cambodia, Venezuela, and Jordan. They account for over 90% of autocracies' total GDP and population. Only China is equal to 14 Vietnams or 10 Russias or four U.S. in population. The spill-over effect is non-negligible: they significantly trade with non-WTO autocracies (Applebaum 2024). For example, Russia, China, or Saudi Arabia can more freely trade with Iran, Iraq, Cuba, and North Korea, while China's post-WTO rapid growth greatly contributed to the commodity boom during the 2000/10s, which benefited non-WTO autocracies (Hamilton 2009; Kilian and Hicks 2012).

Additionally, for semi-engagement, some non-WTO autocracies have been granted MFN status by the United States or the EU, such as Azerbaijan, Belarus, Serbia, and Turkmenistan. Others enjoy varied regional trade deals. Almost all remaining countries were granted WTO observer status.[32]

The Baseline: Gravity Model

I start with the widely used gravity model, regarded as "one of the most robust empirical findings (Chaney 2018)," while absorbing recent methodological improvements (Carnegie 2014; Eaton and Kortum 2012; Goldstein et al. 2007; Rose 2004; Yu 2010). Compared to the matching method below, the gravity framework accommodates continuous Polity scores and exploits bilateral variation across dyad-years to estimate how regime type moderates trade flows on average.

I first estimate the interaction effect of $WTO \times Polity$.[33] WTO membership of exporters

[31] Russia's Polity = 3 in 1995.

[32] Observers must start negotiations within five years, implying they are trying to meet conditionalities and enjoying multiple benefits, such as speaking and learning opportunities as well as signals to investors.

[33] Since post-1990 regime types vary significantly less for most countries (see Appendix), I use Polity in 2000. Alternative measures such as Polity average (2000-20) and real-time Polity nonetheless show consistent results. For pre-1990 years, only real-time Polity is used.

or importers should increase bilateral exports in similar directions, which sum up to aggregate patterns.[34] I specify various gravity models, including: 1) conventionally used WTO dummies *Both_WTO* and *One_WTO* in relation to the reference *Neither_WTO* (e.g., Goldstein et al. 2007; Rose 2004), 2) *Both_WTO* and *One_WTO* (exporter), 3) exporter WTO dummy, and 4) sequential WTO lagged dummies (by three years). I control for a standard set of dyad-level covariates (see Table C.4) and an augmented set of exporter, importer-year, directed-dyad, and year fixed-effects, accounting for "multilateral resistance" effects (Anderson and van Wincoop 2003). The dependent variable is log(exports + 1) (Carnegie 2014). Trade data is drawn from CEPII's Gravity dataset which aggregates data sources such as IMF DOTS and UN Comtrade with over 1,000,000 dyads across 70 years. Given the data and treatment structure, I conservatively cluster robust standard errors by exporter-year and dyad to account for intra-country and intra-dyad error-term correlations.[35]

	Exports (FE)		Exports (FE)		Exports (FE)		Exports (FE)		Exports (CRE)	
	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990
$One\ WTO$	0.304***	−0.131**								
	(0.052)	(0.061)								
$One\ WTO \times Polity_i$	**-0.003**	**-0.025*** **								
	(0.005)	(0.008)								
$Both\ WTO$	0.615***	0.143	0.314***	0.273***					0.309***	0.209***
	(0.084)	(0.092)	(0.046)	(0.047)					(0.015)	(0.012)
$Both\ WTO \times Polity_i$	**0.007**	**-0.041*** **	**0.011*** **	**-0.033*** **					**0.012** **	**-0.038*** **
	(0.006)	(0.010)	(0.004)	(0.006)					(0.001)	(0.002)
$WTO_i\ Only$			0.306***	−0.037					0.304***	0.099***
			(0.052)	(0.065)					(0.017)	(0.017)
$WTO_i\ Only \times Polity_i$			**0.005**	**-0.037*** **					**0.004** **	**-0.038*** **
			(0.005)	(0.007)					(0.001)	(0.002)
WTO_i					0.307***	0.210***	0.240***	−0.022		
					(0.042)	(0.045)	(0.049)	(0.059)		
$WTO\_i \times Polity_i$					**0.009** **	**-0.033*** **	**-0.001**	**-0.003**		
					(0.004)	(0.006)	(0.006)	(0.008)		
$WTO_i\ (3y\ lag)$							0.105**	0.365***		
							(0.042)	(0.051)		
$WTO_i\ (3y\ lag) \times Polity_i$							**0.012** **	**-0.032*** **		
							(0.005)	(0.007)		
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Exporter Means									✓	✓
Dyad Means									✓	✓
Exporter FE	✓	✓	✓	✓	✓	✓	✓	✓	RE	RE
Importer-Year FE	✓	✓	✓	✓	✓	✓	✓	✓		
Dyad FE	✓	✓	✓	✓	✓	✓	✓	✓	RE	RE
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num.Obs.	210 809	538 470	210 809	538 470	210 809	538 470	208 242	502 944	210 049	521 997
R2 Adj.	0.858	0.886	0.868	0.886	0.868	0.886	0.869	0.892	0.874	0.882
BIC	805 515.9	2 258 339.3	833 195.6	2 258 125.3	833 258.2	2 258 320.9	824 099.5	2 103 706.2	691 636.8	1 901 073.2

* p < 0.1, ** p < 0.05, *** p < 0.01

Table 3: The Effects of Joining the WTO. *Note*: Full specifications in Table C.4. Robust standard errors are clustered by exporter-year and dyad for FE models. Institutions such as PR protection is not included to avoid possible post-treatment bias of the WTO. Nonetheless, results hold with inclusion.

[34] E.g., by influencing contract enforcement, institutional reliability, technology and management knowhow transfer, and trade costs, as well as importers' trade barriers and trust in importing from exporter.

[35] I avoid over-clustering by country as error patterns for the same country in the 1990s may be much different from the 2010s (Yu 2010).

Table 3 displays the results of various operationalizations of WTO membership. By looking at the $WTO \times Polity$ interaction terms (in bold), the effect of the WTO on exports is larger for democracies pre-1990, but larger for autocracies post-1990.[36] For magnitude, WTO membership on average inflates exports by 13.9% for average autocracy (Polity = -5), while decreasing exports by 16.4% for average democracy (Polity = 5). One explanation for the negative effect for democracies may be that trade opening-up is subject to more import competition. The results are robust to importer type differentiation (being prior-1990 joiners, regime type). Figure 7 visualizes the rolling estimates for WTO membership.

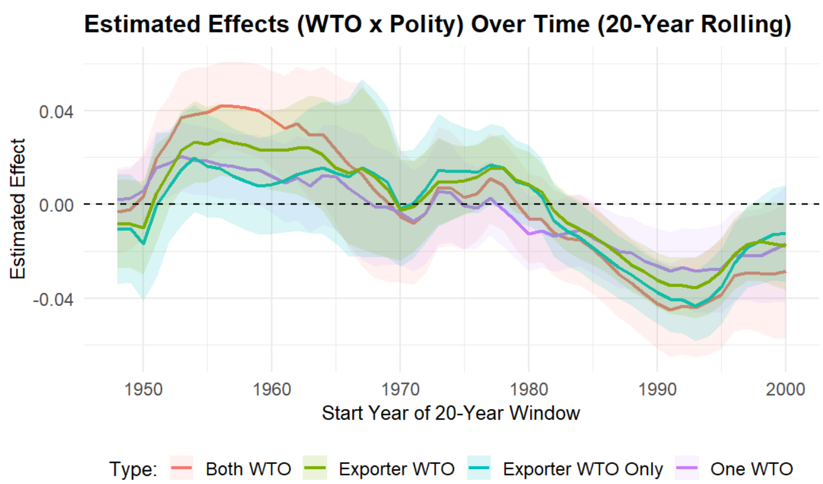


Figure 7: Rolling Estimates of Various Types of WTO Effects Moderated by Polity. *Note*: The effects are estimated by gravity models above within 20-year rolling windows from 1948 to 2020. For example, 1980 denotes the window 1980-2000.

I conduct sensitivity tests following Cinelli and Hazlett (2020) gauge how strong an omitted confounder needs to be to completely explain away the effect of, i.e., $WTO \times Polity$. Figure C.9 plots the sensitivity contours which represent the estimates of $WTO \times Polity$ given the hypothetical partial R^2 of the omitted confounder with treatment ($R^2_{D\sim Z|\mathbf{X}}$) and outcome ($R^2_{Y\sim Z|D,\mathbf{X}}$). Any omitted confounder that nullifies the estimates would need to be over 250 times and 1000 times as strong as GDP and population with both treatment and outcome, respectively.[37] Additionally, I estimate a

[36]The post-1990 interaction effect is moderately smaller if removing China-, or Russia-, or thirteen OPEC-origin dyads, but becomes close to zero if all are removed, which is nonetheless better than the pre-1990 negative effect. Note that removing all that account for the majority of autocracies' GDP (over 90% in 2015) raises representation bias. More importantly, removing all doesn't affect subsequent tests of "engaged reformer."

[37]As noted by Cinelli and Hazlett, these results are conservative for the case of multiple (possibly non-linear) omitted confounders.

hierarchical correlated random-effect (CRE) model with exporter and dyad random-effects, controlling for covariates as well as the means of covariates. CRE allows arbitrary correlations between the exporter-specific and dyad-specific intercepts and predictors to mitigate the assumption violations. As shown in Column 9&10 of Table 3, the result is similar to the fixed-effects model.

Autocratic Engaged Reformers

As argued in Section 4, the effect of WTO membership moderated by regime type is conditional on domestic institutions. When institutional levels are too low or too high, I don't expect more autocratic regimes to significantly outperform others.[38] I estimate the moderated effects stratified by levels of rule of law and PR protection, respectively. Institutional levels are divided into three categories: low, mid, and high.[39] I measure institutional levels by "10-year average institutional levels" after accession and assign dyads into the corresponding institutional categories. The idea is to test how the regime-moderated WTO effect varies across institutional levels.[40] Since autocratic advantages amplify exports most at moderate levels, I combine low and high coded as other. I thus define reformed as (0.2-0.7) in the rule-of-law case, for example; autocracies are empirically absent above 0.7, establishing the upper bound of common support. Results are robust to binary categories (low, above) and excluding democracies above 0.7 entirely (Appendix). I fit a model including a three-way interaction among post-1990 joiners:

$$Export_{ijt} = \beta WTO_{it} \times Polity_i \times Instituion10year_i + \delta \mathbf{X}_{ijt} + \gamma_{ij} + \eta_t + \epsilon_{ijt}$$

where $Instituion10year_i$ is a categorical variable of post-WTO 10-year averages of institutional levels (low, mid, high) of country i, and $\mathbf{X}_{ijt}$ denotes dyad-level covariates. γ_{ij} and η_t are dyad and year fixed-effects.

[38] Note that the theory applies to the continuous Polity measure.

[39] I calculate thresholds combining lowest/highest 20 percentiles of institutions among developing countries in 2000 with minor adjustment based on real-world cases (see Appendix for details). The final ranges are (0.2, 0.7) for rule of law and (0.35, 0.85) for PR protection, both retrieved from VDem.

[40] I do not control for institution in the model for possible post-treatment bias: institutional change may be partly affected by WTO membership. Yet, this may neglect pre-WTO institution's effect. However, controlling for institution doesn't affect results.

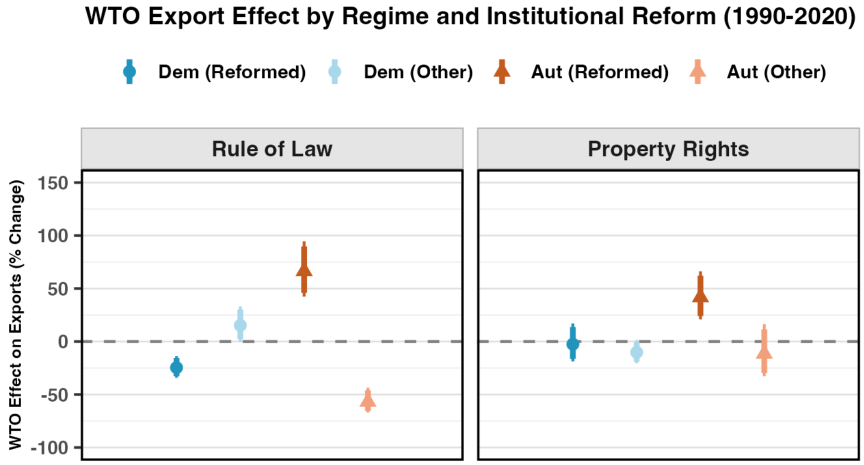


Figure 8: The Effects of Joining the WTO Conditional on Institutional Levels. *Note*: The y-axis means the difference of WTO effects across Polity. A positive value means the WTO favors democracy.

Figure 8 shows that, as expected, autocratic reformers outperform all other regime categories. For example, WTO accession increases their average exports by nearly 60% and 40% for rule of law and PR protection, respectively. Robustness tests include varying institutional thresholds up/down 5 percentage points across a range of +/-10 (Figure 9), changing post-WTO institution averages to 5 or 15 years, and using dichotomous Polity (see Appendix). I also run: (1) models with both-WTO variable (treatment at dyad-level) and (2) a correlated random-effects model, all of which yield similar results (see full table in Table C.5).

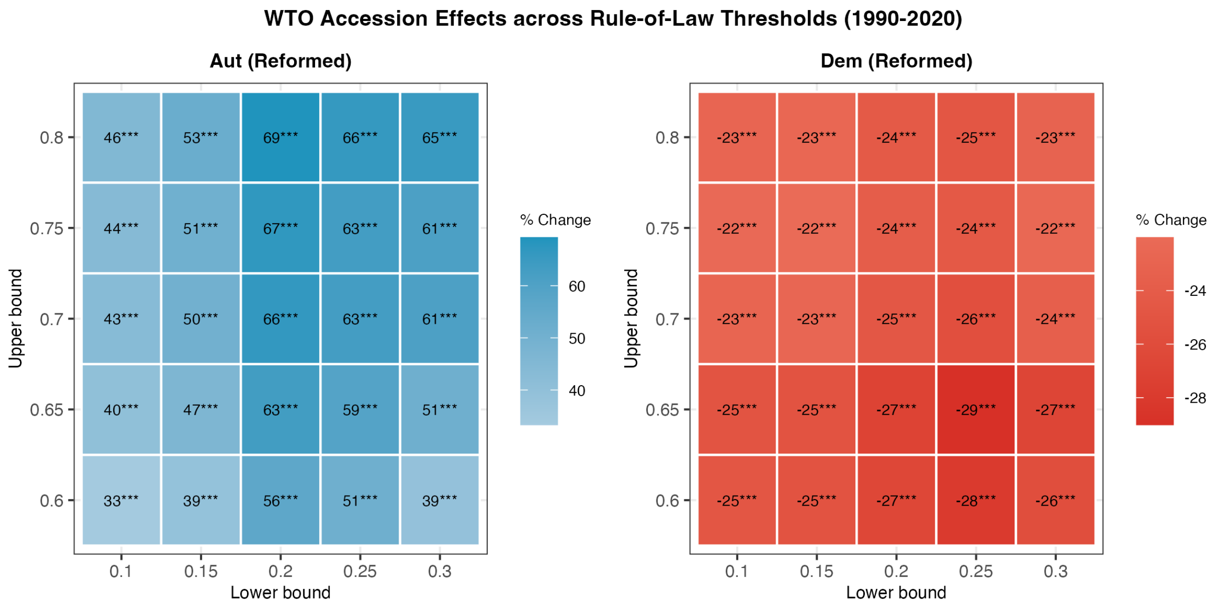


Figure 9: WTO Accession Effects across Rule-of-Law Thresholds (1990–2020). *Note*: Each cell reports the estimated percentage change in exports from WTO accession under alternative rule-of-law classification cutoffs. The lower (upper) bound defines the maximum (minimum) institutional levels. Baseline: lower = 0.2, upper = 0.7. *** $p < 0.01$.

The most significant challenge is the selection of autocratic engaged reformers who export more. The empirical models address this directly by controlling for numerous pre-treatment covariates such as income levels and institutional quality. Moreover, the central comparison is between autocratic and democratic engaged reformers both of which selected into the same scope condition—reform and integration—suggesting roughly similar baselines of unobservables driving that selection. The residual gap—within a group selected on the same criterion—is what regime type predicts.

The reason that no pre-1990 autocratic advantages were shown is that there had been neither substantive market-oriented reforms worldwide (or only on paper for some newly decolonized states) nor globalization. In other words, the scope condition had not been sufficiently met. While some autocracies and democracies did trade before 1990, the scale, depth, and institutional environment were categorically different from the globalized economy. The global value chain had not taken off—for example, South Korea largely relied on developing indigenous supply chains (Baldwin 2016).

Staggered Difference-in-Differences (DiD) with Weighting

In addition to model-based approaches, I use staggered DiD with matching as a nonparametric identification strategy to estimate the effect of WTO accession. Apart from directly testing country rather than dyad levels, it offers significant advantages over, e.g., fixed-effects methods for panel data (Imai et al. 2022), by explicitly constructing counterfactuals through matching on pre-treatment covariates.[41] This approach ensures comparable comparisons and reduces outlier influence, providing more robust causal estimates. It also provides insight into the long-term effects, as the WTO effect may persist over time. The ATT (Average Treatment effect on the Treated) estimator is expressed as follows:

$$\hat{\tau}^{ATT} = \frac{1}{\sum_{i=1}^{N}\sum_{t=L+1}^{T-F} D_{it}} \sum_{i=1}^{N}\sum_{t=L+1}^{T-F} D_{it} \left\{ \left(Y_{i,t+F} - Y_{i,t-1}\right) - \sum_{i' \in \mathcal{M}_{it}} w_{it}^{i'} \left(Y_{i',t+F} - Y_{i',t-1}\right) \right\}$$

where D_{it} is treatment indicator (1 at treatment onset). $Y_{i,t}$ is the outcome for treated unit i at time t. $\mathcal{M}_{it}$ is the set of matched control units for treated unit i at time t. $w_{it}^{i'}$ is the weight for control unit i' matched to treated unit i.

Specifically, Covariate Balancing Propensity Score (CBPS) weighting is used to balance covariates.[42] CBPS estimates propensity scores such that covariates are balanced (Imai and Ratkovic 2015). Weighting methods are particularly effective in non-large datasets because they retain all available control units. As DiD is inconvenient for handling interaction effects, units are roughly stratified into democracies (Polity ≥ 0) and autocracies (Polity ≤ 0).[43] For all tests, I use export (log) as the DV, similar to gravity models.

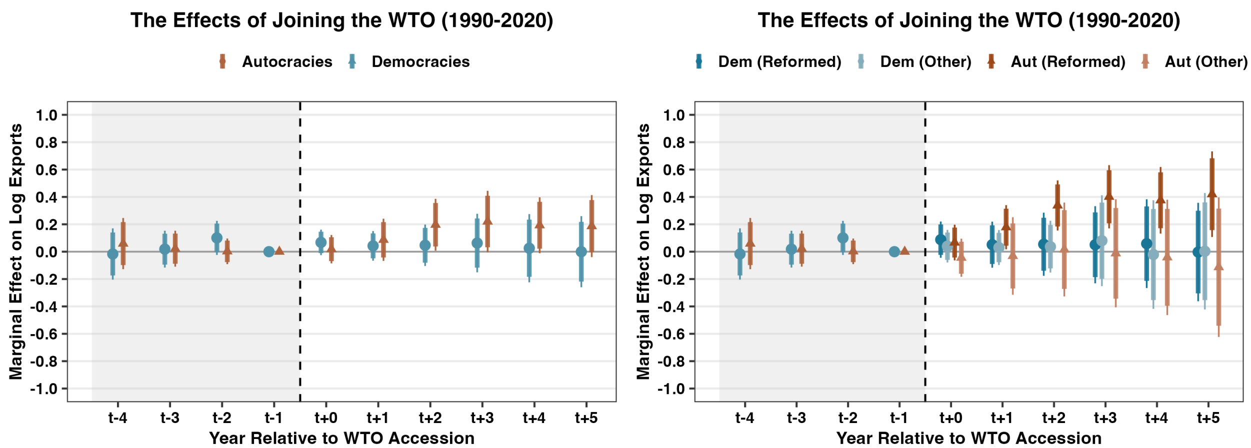
Utilizing the country-year panel dataset, I set lags $L = 4$ to match pre-treatment histories and

[41] PanelMatch is better suited to this setting than Callaway and Sant'Anna (2021) due to thin cohorts and possible pre-treatment imbalances.

[42] I choose among mahalanobis matching, propensity score matching/weighting, and CBPS matching/weighting for the best performance in balancing covariates. The standardized mean differences (SMD) of most covariates are within the rule-of-thumb of 0.2 (see Figure C.10).

[43] As regime types have remained relatively stable since the mid-1990s, I capture regime type in 2000 for the same country throughout the period.

$F = 5$ for forward effects, as joining the WTO may not immediately boost trade.[44] I focus on pre-treatment covariates that theoretically affect both WTO accession and exports, including GDP (log), GDP per capita (log), Polity, population (log), race (white), geopolitics (NATO membership), natural resource intensity (% of GDP), industrial output intensity (% of GDP), rule of law, and lagged exports (for pre-WTO export-level self-selection). I avoid controlling for direct post-treatment covariates such as tariff rate, which is lowered upon WTO accession.[45]



a. by Regime Type

b. by Regime Type and Reform Status

Figure 10: Effects of Joining the WTO on Exports (Post-1990). *Note*: Autocracy: Polity ≤ 0. The WTO effect of autocracy (and engaged reformer) is statistically significant while those of others are not. Placebo periods show that parallel trends assumption hold (the shaded area, with t-1 as reference time). Standard errors are estimated with bootstrap.

Figure 10 plots the WTO effects for democracy and autocracy, respectively. Panel a shows that from t+3 to t+5, autocracy's WTO effect is approximately 0.2, while democracy's is insignificant. This is consistent with gravity models (Table 3), which report that the effect difference between average autocracy (Polity = -5) and average democracy (Polity = 5) is approximately 0.3 for the post-1990 period. Panel b reports that autocracy's effect is primarily driven by autocratic engaged reformers.[46]

In Figure 11, I show China (the largest autocracy), Angola (a resource-rich African autocracy), Ukraine (a post-Soviet democracy), and Costa Rica (a development success in Latin America) as

[44] Longer leads and lags are refrained from, as they can eliminate more units that don't match.

[45] I match rule of law because WTO conditionality (post-treated) is more about trade-related liberalization and intellectual PR protection (Allee and Scalera 2012). Results hold for no matching.

[46] Same as above, rule of law between 0.2 and 0.7.

examples using the synthetic control method (SCM). China, for example, compared to the control unit, had 189% and 242% more exports 10 years and 15 years after WTO accession, respectively.

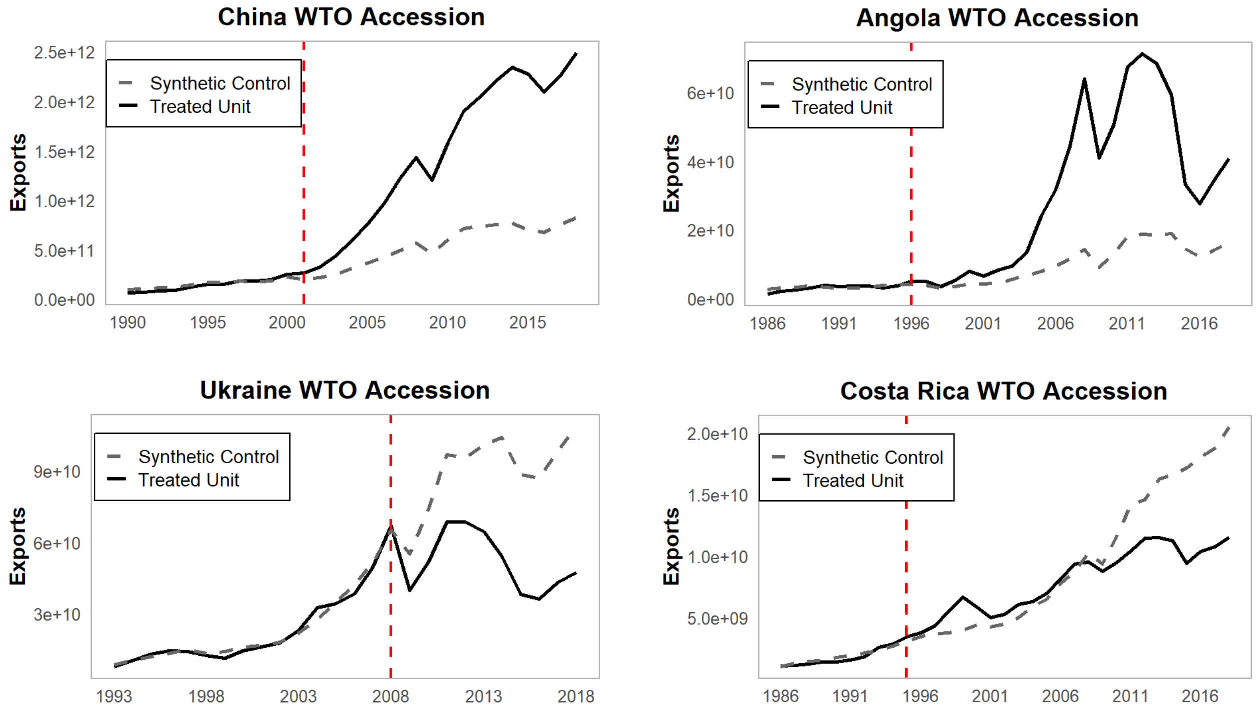


Figure 11: Examples by Synthetic Control Method. *Note*: Red dashed lines denote WTO-joining years. Covariates to predict the control unit include: GDP, GDP per capita, GDP growth, population, trade openness, Polity, PR protection, resource intensity, industrial intensity, and lagged exports.

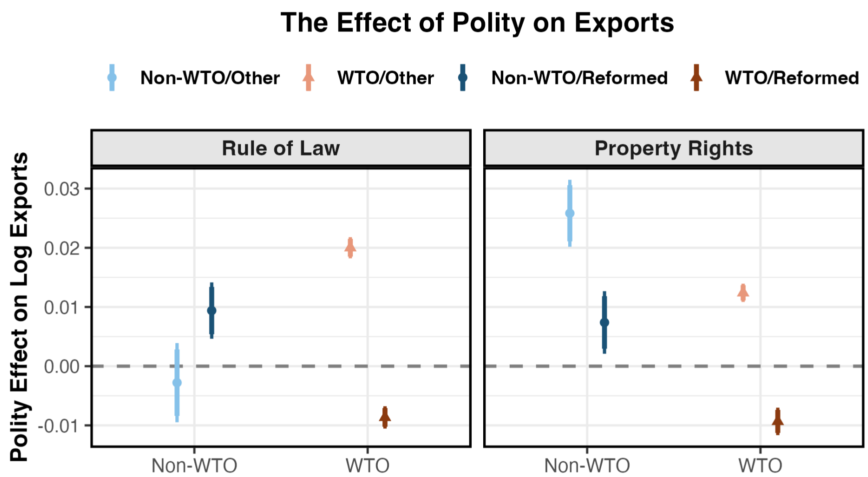
This contrast suggests that autocracies likely have much weaker export performance absent trade engagement, because autocratic institutions usually suppress domestic demand making external demand competition essential to their growth model. Trade isolation forecloses precisely this opportunity. In Appendix E.1, I show that domestic reform alone yields much weaker effects during their non-WTO periods.

5.2 Absolute Levels of Trade Performance

To test whether the regime category of autocratic engaged reformer leads to higher absolute export levels under a globalized economy, I estimate full gravity models on the pooled dyadic sample (169 countries, 532,574 dyad-years over 1990-2020) and divide all dyads into four regime categories (WTO $\times$ Reformed/Other) based on exporter.[47] Models are estimated separately within each cell

[47] I use the same institutional categories as the above tests for rule of law and PR protection (lagged).

by weighted least squares with inverse-density weights.



(a)

Figure 12: Polity Effect on Exports by WTO Status and Institutional Quality. *Note*: Coefficients from gravity regressions of log bilateral exports on polity score, estimated separately for WTO/Non-WTO countries and institutional reform status.

Figure 12 reveals a consistent pattern across both institutional measures. Among WTO members with moderate institutional levels (WTO/Reformed), being more autocratic is associated with significantly higher exports—the polity coefficient is negative. This autocratic advantage does not extend to WTO members outside the moderate range (WTO/Other) where the polity effect turns positive, nor to non-WTO reformed countries (Non-WTO/Reformed) where democracy favors exports. The results thus confirm the theoretical expectation: the export premium of autocracy is specific to countries that combine WTO membership with moderate institutional reform, which are the defining characteristics of autocratic engaged reformers.

6 Mechanism Evidence

What drives the export premium for reformed autocracies? I provide three types of mechanism evidence: sector heterogeneity tests examining whether the effect is concentrated where the mechanisms predict, case evidence from China and India tracing governance features to export policy, and investor perception analysis drawing on Japanese financial news and Japanese firm surveys.

6.1 Sectoral-level Heterogeneity

If autocratic advantages enhance exports through broad institutional features such as centralized and coordinated mercantilist policies or active/passive wage suppression, they should favor exports of manufactured and commodity goods, as well as intermediate goods that reflect GVC participation, compared to agricultural and primary goods. For example, China shifted millions of laborers from agricultural sectors to manufacturing after WTO accession (Erten and Leight 2019).

The UNCTAD classifies sectors based on manufacturing factors (labor, natural resource, and technology).[48] Accordingly, I create four categories of product types: agriculture, natural resource, basic manufacturing, and mid-to-high manufacturing based on manufacture types at the broad Harmonized System (HS2) level. The World Bank's WITS on the other hand, classifies Harmonized System sectors into raw materials, intermediate goods, consumer goods, and capital goods based on GVC participation. I merge six-digit HS code into the four broad categories. I then ran sector-level gravity models with the interaction term (WTO $\times$ Reformed/Other) using CEPII's BACI data.

Figure 13 presents consistent results across both product classifications. Reformed autocracies show large positive effects in every sector, from roughly 65% in agriculture to over 185% in capital goods—while all other groups cluster near zero, regardless of sector. The advantage is absent for unreformed autocracies and democracies alike, confirming that performance is specific to the intersection of regime type and domestic reform.

[48]See https://unctadstat.unctad.org/EN/Classifications/DimSitcRev3Products_Tdr_Hierarchy.pdf.

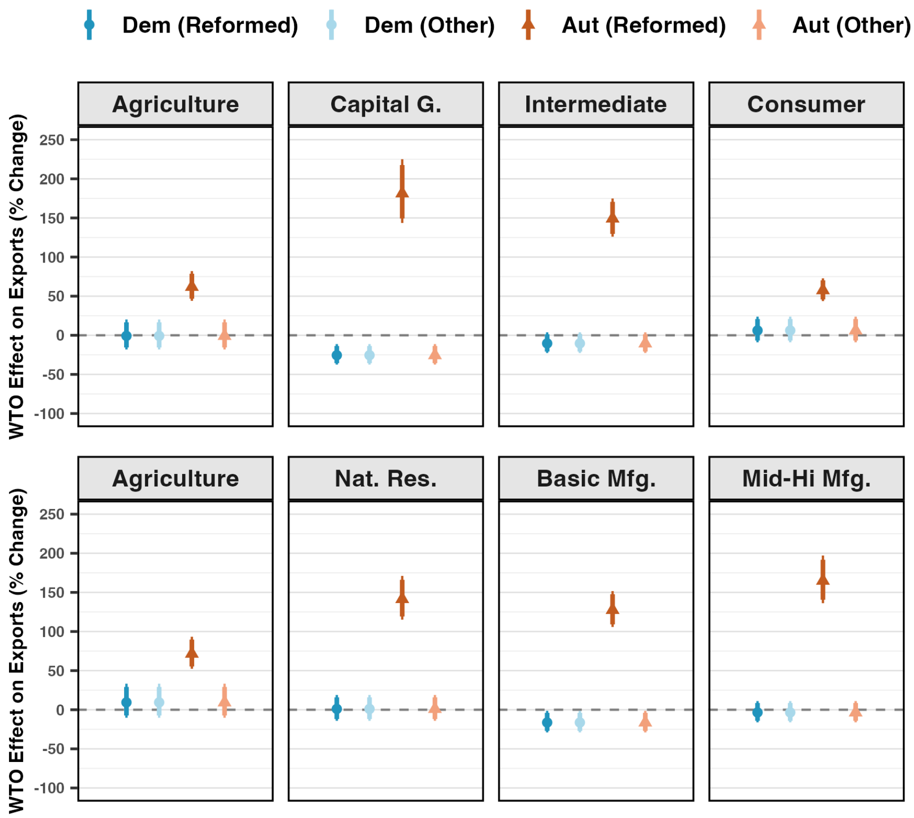


Figure 13: WTO Effect across Sectoral Categories (1990-2020). *Note*: Estimated WTO accession effects on bilateral exports (% change) across four regime-reform groups and two HS2 sector classifications. Gravity models with dyad, dyad-sector, and year fixed-effects; standard errors clustered at the dyad level.

The sector-level pattern offers additional leverage. Gains are comparatively smaller in agriculture and consumer goods, where WTO obligations are weakest and state coordination is least decisive. The largest effects appear in capital goods, intermediates, and mid-to-high manufacturing, consistent with FDI-driven upgrading and coordinated industrial policy (echoing the descriptive comparisons in Table 2), rather than labor cost competition alone. The effect, moreover, extends well beyond natural resources, effectively ruling out a commodity-driven explanation. Results are robust when Polity is treated as continuous: the WTO export effect declines monotonically as regime type becomes more democratic across all sector types (Appendix).

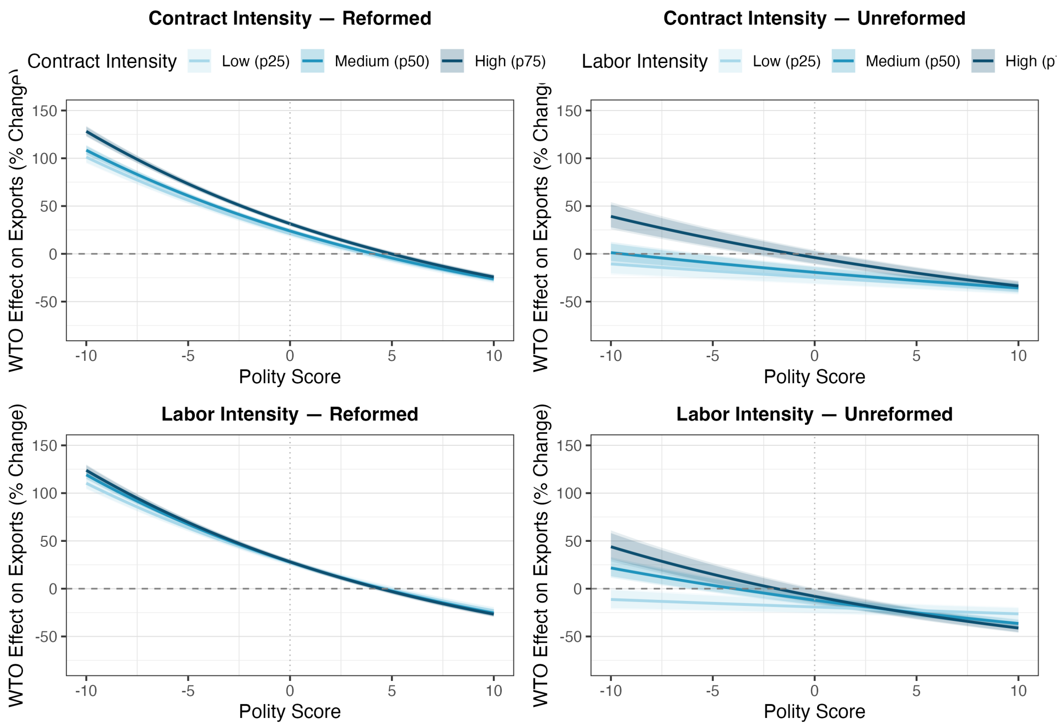


Figure 14: Contract and Labor Intensity Heterogeneity in WTO Effects (1990-2020). *Note*: WTO accession effect on bilateral exports across Polity scores at the 25th, 50th, and 75th percentiles of contract intensity (Nunn 2007) and labor intensity (NBER-CES), measured at HS2 level. Gravity models with dyad-HS2 and year fixed effects; standard errors clustered at dyad-HS2.

Figure 14 plots the WTO effect across Polity scores at low, medium, and high levels of contractibility (Nunn 2007) and labor intensity (NBER-CES) for reformed and unreformed countries separately. For reformed joiners, contract intensity favors autocracies and higher level generates a positive premium, consistent with state coordination in relationship-specific sectors (Nunn 2007). The effect by labor intensity is larger for autocracies, but shows no differentiation—undermining a pure wage-suppression account. Unreformed autocracies show much smaller overall effects with modest labor-intensity differentiation, suggesting their gains are more narrowly labor-cost driven. Their contract-intensity premium may reflect coercive enforcement capacity or simply their resource-heavy sectors that score high on contract intensity.

6.2 Case Study: China vs. India [to extend]

To understand the proposed within-state micro-level mechanisms, I compare China and India, the largest autocracy and democracy, respectively. China and India, though not identical, share many structural similarities prior to the early 1980s: demographics, trade-related geography such as coastal proximity, resource endowment, and an inward-looking, heavily planned economy. India, while not strictly limiting private ownership like China did, maintained an extensive system (License Raj) of licenses, permits, and regulations that controlled the private sector since the 1950s. For decades, both countries unsurprisingly had similar trajectories in export performance (Figure 15), but they started to diverge once China became an "engaged reformer."

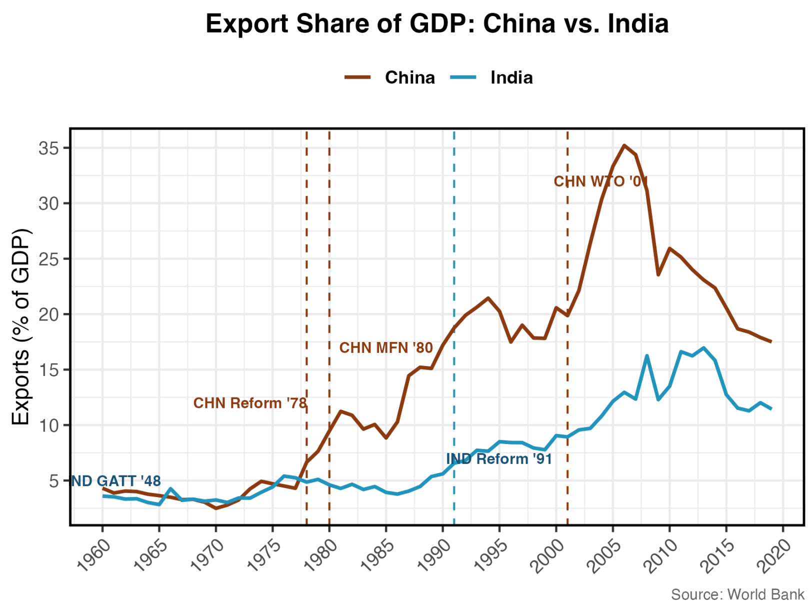


Figure 15: Exports (%): China and India.

China fits my theory particularly well: initiating market-oriented reforms in 1978, it secured MFN status from the U.S (i.e., semi-engagement) in 1980 while borrowing hugely from the World Bank. Though still recording persistent trade deficits (1980-1995), China's exports immediately took off. China's WTO accession in 2001 provided another export boost, leading to sustained trade surpluses thereafter. Specifically, China's authoritarian regime enabled features such as a gradualist, coordinated, and rapidly responsive approach blending state control with markets and long-term planning (Lin et al. 2003), mercantilist policies, and labor rights suppression. By contrast, India's GATT signatory status back in 1948 didn't stimulate trade much. Despite having a higher nominal PR protection than China (0.77 vs. 0.35 in 2000), India's democratic system, though more inclusive, must navigate coalition politics, public dissent, and legal constraints, resulting in slow decision-making and limited mercantilist practices (Groth 2006). Consequently, India's export performance diverged from China's around 1980, running persistent trade deficits. However, conforming to the theory, with Modi's more centralized and authoritarian turn, India has quickly pivoted toward a more mercantilist trade policy, particularly in electronics.[49] In only five years (2018-2023), India's electronics exports increased fivefold to $22 billion.

6.3 Investor Perception: Japanese Firms in Vietnam vs. the Philippines

Global firms' investing preferences are key to national development in the post-1990 GVC era. I examine Japanese investors' preferences over Vietnam and the Philippines, which present another instructive pair comparison. In the 1980s, both countries shared similar income levels (both below $600), economic structures, demographic profiles, geographic conditions, ethnic compositions, and resource endowments.

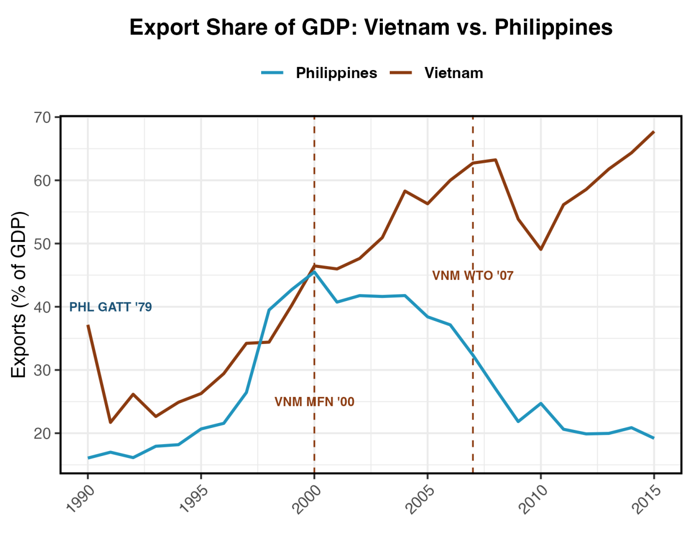


Figure 16: Exports (%): Vietnam and the Philippines.

Vietnam initiated economic reforms in 1986, and by 2001 had obtained MFN status from most Western countries. Although running persistent trade deficits before 2010, Vietnam had implemented a state-led, mercantilist development strategy modeled in part on China's. The country's ability to pursue consistent policy agendas, long-term planning, and gradual and managed liberalization has been widely attributed to its centralized political system (Kirkpatrick et al. 2001).

[49] "10 Years of Make in India," Government of India, 2024.

Vietnam's WTO accession in 2007 marked a turning point, and within five years, it ran sustained trade surpluses. In contrast, the more democratic Philippines witnessed elites dominating the democratic process and capturing rents, while resources were diverted away from investment in human development and infrastructure, key to export performance (Baulch 2016).

Table 4: Japanese Affiliate Survey Evidence: Vietnam vs. Philippines, 2009–2019

Metric	**Country**	**2009**	**2011**	**2013**	**2015**	**2017**	**2019**
Panel A: Japanese FDI Presence (firms in JETRO database)							
Firms in JETRO database	Vietnam	277	292	585	1,027	1,345	1,608
	Philippines	333	230	237	239	382	517
Panel B: Survey Responses (% of responding firms; N in parentheses)							
Expansion intent (%)	Vietnam	77.6 (76)	68.0 (150)	~65 (435)	63.9 (557)	69.5 (652)	63.9 (855)
	Philippines	<50 (115)	43.0 (121)	~55 (150)	55.1 (119)	63.4 (73)	51.8 (139)
Wage hike: top challenge (%) [rank among problems]	Vietnam	85.2 [#1]	83.3 [#2]	78.2 [#1]	77.9 [#1]	75.2 [#1]	72.0 [#5]
	Philippines	<top 3	48.4	41.8 [#5]	31.4	45.8 [#5]	44.5
Production (goods) as expansion function (%)	Vietnam	—	—	38.3 [#2]	42.4 [#1]	35.9 [#2]	—
	Philippines	—	—	30.6	23.1	16.3	—
BPO/Admin services as expansion function (%)	Vietnam	—	—	low	low	low	—
	Philippines	—	—	high	44.4 [#1]	44.4 [#1]	—

Source: JETRO *Survey of Japanese-Affiliated Firms in Asia and Oceania*, 2009–2019. Panel A counts all Japanese-affiliated firms (≥10% Japanese equity) registered in JETRO's database; Panel B percentages are computed over responding firms only. ~ = approximate; — = not disaggregated.

Table 4 collects and summarizes six rounds of data based on JETRO's annual "Survey of Japanese-Affiliated Firms in Asia and Oceania" to provide firm-level evidence that investor perceptions translate into actual investment behavior.[50] The aggregate export patterns in Section 4 leave open whether firms consciously respond to the governance mechanisms theorized here. The JETRO data provides partial answers. Panel A shows Japanese firm presence in Vietnam growing nearly sixfold between 2009 and 2019, while Philippine presence barely grows. Panel B reveals two patterns consistent with the mechanisms: Vietnam's firms report persistently higher expansion intent, and wage hikes rank as the top operational challenge as foreign firms pour in—suggesting that labor costs have been suppressed and contributed to Vietnam's competitive position—while Vietnamese expansion is concentrated in goods production rather than the services and BPO functions that dominate Philippine operations, consistent with Vietnam's mercantilist, manufacturing-oriented investment climate.

[50]JETRO (Japan External Trade Organization) is a government-affiliated agency that annually surveys Japanese-affiliated firms.

To test whether the four mechanisms in Section 3 manifest in investor discourse, I conduct a systematic content analysis of Japanese financial news coverage of investing in Vietnam and the Philippines. I collect keyword-filtered articles from the Nikkei, Daily Yomiuri, and related outlets via Factiva spanning 2001-2020, yielding over 10,000 articles coded as relevant to investment in either country. Each article is scored by GPT-4o-mini on five dimensions corresponding to four theoretical mechanisms—centralized power, weak institutional constraints, limited accountability, mercantilist mentality, plus overall attractiveness—using both a continuous salience scale (1-5) and a binary presence indicator (0/1). To validate LLM coding reliability, I independently code a random subsample of 100 articles, yielding a high inter-rater agreement. I then estimate two OLS models regressing each dimension score on a Vietnam indicator, with year and outlet fixed-effects, and log article length as controls with robust standard errors.

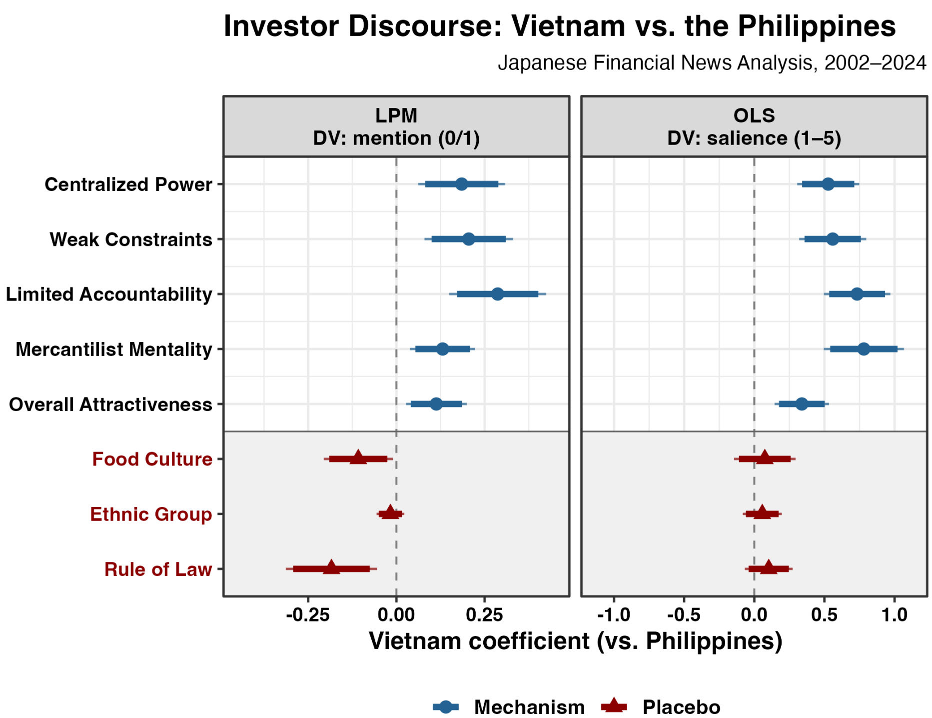


Figure 17: Investor Perception Analysis. *Note*: data retrieved from the articles of major Japanese Financial Newspapers. Scores are rated by LLM (GPT-4o) with human validation.

In Figure 17, Vietnamese articles score significantly higher than Philippine articles on all five dimensions across both the continuous and binary specifications. For example, the limited accountability mechanism—proxied by labor cost competitiveness and suppressed labor rights—

where Vietnam articles score 0.73 points higher on the salience scale (panel b) and are 28.7 percentage points more likely to contain this framing (panel a). To assess whether the results reflect investment-relevant perceptions rather than general country-specific bias, I add three placebo dimensions: food culture, ethnic group, and rule of law. The result shows no systematic Vietnam advantage for placebos.

7 Alternative Explanations

Catching up

Less developed countries (LDCs) may converge with developed economies through faster growth (Luo and Przeworski 2019). Yet, post-1990 WTO-joiners were mostly LDCs. Neither did autocracies start low. The correlation between GDP per capita (pre-WTO accession or in 1990) and Polity for post-1990 WTO-joiners is negative (r = -0.25 and -0.43)—autocracies started higher. Absolute trade levels also favor autocracies controlling for income per capita. Moreover, catch-up cannot predict why reformed and non-reformed autocracies move in opposite directions.

Importing Credibility

Arias et al. (2018) argue that autocracies benefit more from joining international institutions since they rely more on external credibility. Importing institutions is one way to increase performance. However, this theory cannot explain why reformed and other autocratic WTO members perform differently. It is the combination of institutional reform, economic engagement, and autocratic institutional advantages that matters. Historically, domestic institutional improvement results much from active domestic reform pushed by governments.

Other Discussions

Trade integration and institutional reforms are arguably the two most important determinants of trade performance identified in the growth and development literature. Nonetheless, below I discuss some potential questions.

Is it simply a story of China, Vietnam, Russia, and oil states? None of the descriptive data in Section 2 shows they are outliers. Moreover, the theory and empirical evidence of engaged reformer suggest that regime type plays an important role. Even if one insists that the theory fits better with the above countries, which are major autocracies, it largely explains autocratic rise.

What about the commodity boom and the spillover effect? I first delete the boom years (2004-2014) and OPEC countries, and the results hold. The spillover of joiners (e.g., China and others) and the buildup of a global commodity market do matter. However, this second-order effect does not negate my argument that globalization facilitates autocratic rise. Plus, we do not observe autocratic advantages in the previous oil boom (1970s/80s), and resource-rich countries that do not meet the scope condition nonetheless underperformed (e.g., Venezuela, Iran, and Iraq). Gulf successes may be less remarkable absent structural changes in globalized commodity markets, and globalization-driven demand, investments, and technology diffusion. In fact, oil industries are capital- and technology-intensive.

State capacity plays a pivotal role in economic development (Acemoglu et al. 2015; Dincecco 2017). Yet the association between high-performing states and strong capacity does not necessarily imply causation, that is, conditional probability $P(\text{capacity} \mid \text{performance})$ is not $P(\text{performance} \mid \text{capacity})$. Some high-capacity states, such as North Korea and Iran, have nevertheless under-perform. In the Appendix, I examines the robustness of the results to broader measures of state capacity.

For semi-engagement such as prior MFN status, some are granted as part of PTAs, for example, U.S.-Vietnam or U.S.-Lao Bilateral Trade Agreements, which are controlled for in the models. Moreover, MFN is a core WTO principle and semi-engagement is part of globalization, while WTO membership provides much more benefits than a revocable MFN status. More importantly, any export gains from pre-accession MFN are absorbed into pre-treatment trends, meaning the estimated WTO effect is likely a lower bound on the full effect of institutional trade engagement.

Back-and-forth trade along the GVC may double count exports. However, democracies are usually more GVC-integrated, their gross exports may overstate true trade performance relative to autocracies. Moreover, external balance—net exports that net out intermediate inputs—also favors autocracies. Gross exports also remain the conventional benchmark for trade performance.

Lastly, although not the focus of the paper, export-oriented FDI, rather than service-oriented, can boost exports (Helpman 1984; Pandya 2016). As export-oriented FDI usually follows globalized production (Helpman 1984; Markusen 1984), it is better understood as a post-treatment mediator or channel: it follows trade engagement rather than preceding it (Carnegie 2014). While the greenfield FDI data (fDi Markets) is only available since 2003, the pattern of higher inflows among autocratic engaged reformers is consistent with the mechanism (Table 2).

8 Conclusion

The debate over regime type and economic performance has long hinged on the mechanisms through which political institutions shape development. This paper advances that debate by addressing a specific puzzle: why has the effect of regime type on trade performance changed so dramatically in the post-1990 period? I argue that economic globalization constitutes a necessary condition for autocratic trade advantages—one that prior scholarship has underweighted. Because autocratic regimes often lack inclusive institutions for growth, resulting in weak domestic demand (Acemoglu and Robinson 2012), they should find growth harder without strong external demand. Leveraging WTO accession as a source of exogenous variation, the empirical findings confirm that autocratic engaged reformers systematically outperform democratic counterparts on export performance—a gap not explained by pre-treatment differences or selection into engagement.

The findings carry several theoretical implications. First, they qualify institutional theories of growth: while inclusive institutions matter for domestic demand-driven development, external demand can substitute for — and even reward — the exclusionary institutional arrangements characteristic of autocracies. Second, they challenge optimistic accounts of trade liberalization. Liberal internationalists have argued that open trade fosters interdependence and democratic convergence; this paper shows that the existing trade regime has created conditions under which autocracies can thrive without political liberalization. Third, the findings speak to Downs et al.'s (2000) accession theory: sequential accession has proven insufficient to ensure compliance with WTO principles, and as autocratic members have thrived, the organization's capacity to function as a rule-based institution has eroded.

The trend depicted in the paper posts a sober future for the current liberal order and democracies.

Some argue that democracies should establish their own trade bloc (Friedberg 2025). Whether framed as a contest between democracy and autocracy or as new great power rivalry, my findings indicate that global market forces – shaped by the existing trade system—may continue to favor autocracies. Given this dynamic, the continued deindustralization of advanced democracies may not be surprising. The fact that economic globalization has produced this outcome is both unexpected and undesirable. Ultimately, where the global trade system and globalization should be headed hinges on its expectations and consequences; as Keohane (1984) posits, the means are justified by the ends. Future research should further explore the institutional advantages of reformed autocracies and whether democracies truly cannot compete on an equal footing.

Should countries embrace autocracy? The autocratic advantage identified here is conditional: it operates only through engagement and reform, and only under conditions of sufficient globalization — as the absence of pre-1990 effects confirms. Should deglobalization deepen through tariff escalation or regional bloc consolidation, the scope conditions enabling these advantages may attenuate. Future research should examine whether democracies face inherent institutional disadvantages in export competition under deep globalization, and whether the political costs of deindustrialization feed back into the democratic erosion that has characterized many advanced economies in recent decades.

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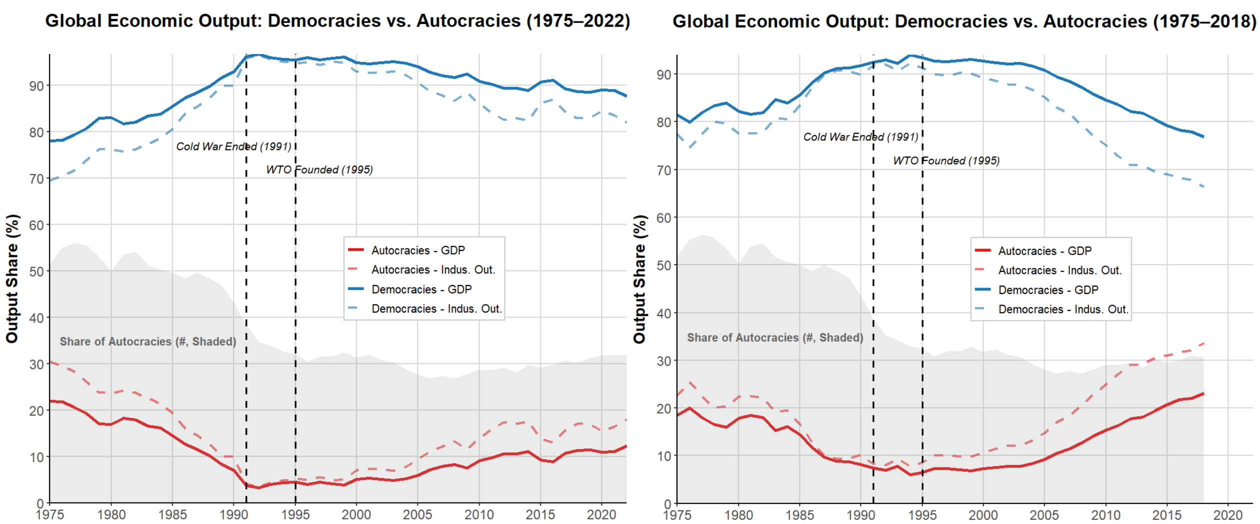
Contents of Appendix

A Data Description

This section details the data that underpin the analysis. Summary statistics, plots, and data distributions are presented to provide a more rich understanding of the story and its suitability for research questions.

A.1 Distributional Change



(a) Freedom House Index (Excl. China)

(b) Polity (All)

Figure A.1: The Distribution of Power Change Between Democracies and Autocracies. *Note*: Data: World Bank. In (a), autocracy is measured by $FH \geq 10$. In (b), autocracy is measured by $Polity \leq 0$. Polity data is available until 2018.

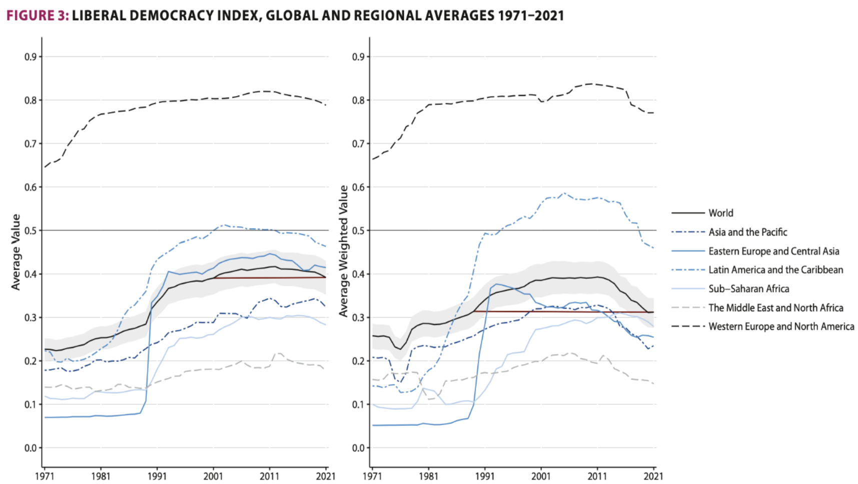


Figure A.2: VDem Global Liberal Democracy Index (Source: the Vdem Report 2021)

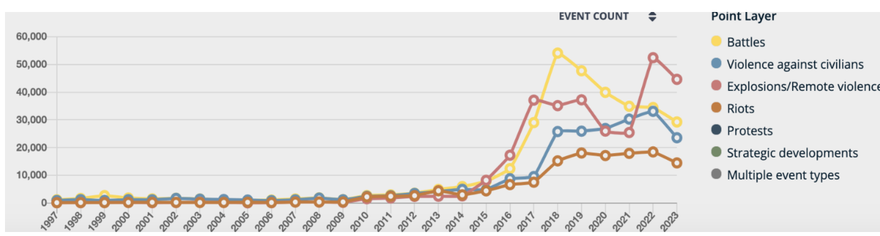


Figure A.3: The Trend of Global Conflicts (Source: ACLED).

Country/Leader	Fast Growth?	Export-Led?	Reason/Details
China (1978–2008)	Yes	Yes	Consistently high growth (10%), driven by market-oriented reform and export-led manufacturing, SEZs, and WTO entry.
Japan (1950s–1970s)	Yes	Yes	Post-war authoritarian, market-minded bureaucracy enabled export-led growth, tech upgrading, and 9% annual GDP growth.
South Korea (1962–1987)	Yes	Yes	Authoritarian state led industrial policy and export push.
Taiwan (1960s–1980s)	Yes	Yes	Under KMT authoritarianism, state-supported export sectors like electronics drove fast growth.
Singapore (1965–1990)	Yes	Yes	Semi-authoritarian regime used strategic export-oriented policy and MNC investment.
Hong Kong (1950s–1980s)	Yes	Yes	Colonial rule with modern market; booming re-export and service economy.
Malaysia (1986–2000)	Yes	Yes	Rapid growth post-1985 crisis; export-oriented industrialization (electronics, palm oil); strong FDI.
Thailand (1986–1996)	Yes	Yes	Export-led manufacturing boom before 1997 crisis; 9% annual growth driven by electronics and autos.
Indonesia (1989–1996)	Yes	Yes	Suharto-era liberalization brought export growth in textiles, natural resources before Asian financial crisis.
Vietnam (1990s–2010)	Yes	Yes	Post-Doi Moi reforms led to 6–7% growth; FDI-driven export economy in garments, electronics, and agri.

Pinochet (Chile, 1980s)	Yes	Yes	7% growth post-1985; market reform; exports quadrupled in 1985-1995
Franco (Spain, 1959–74)	Yes	Partial	average 7% growth rate; authoritarian technocrats with market reforms; GATT 1963, abandoned autarkic policies but on imports and low mercantilist policies; FDI, domestic industrialization mattered more than exports
Junta (Brazil, 1968-80)	Yes	Partial	10%+ growth rate during 'Brazilian Miracle'; market economy; state-led investment in infrastructure and industry and export promotion, exports increased ten-fold in 1968-1980
Salazar (Portugal, 1950/60s)	Yes	Yes	average 5–7% growth rate; market reform; investment in infrastructure and industry; export boom after 1960 EFTA and 1962 GATT, exports tripled in 1960-1970
PRI (Mexico, 1950s-80s)	Yes	Yes	1950-70 state-led industrialization, infra investment, and ISI; market institutions; FDI/tech absorption weaker than Spain; 1970s started export-oriented policies, exports increased thirteen-fold in 1972-82, 1986 GATT and 1994 NAFTA-led export boom.
Fujimori (Peru, 1993-2013)	Partial	Yes	Early ISI not work well; 1980/90s market reforms; post-Fujimori based on early reforms; riding on commodity boom, exports increased twelve-fold in 1993-2013.
dos Santos (Angola, 2002-12)	Yes	Yes	Market reform though limited, attracts foreign oil companies, state-led infrastructure; post-war boom largely driven by oil exports, exports increased by fourteen-fold; .
Ben Ali (Tunisia)	Yes	Yes	Growth via light manufacturing and exports to Europe, especially textiles.
Kagame (Rwanda)	Yes	Partial	7–8% annual growth, with modest export expansion and large public investment.
Botswana (1966–80)	Yes	Yes	Diamond mining accounted for most of Africa's strongest GDP growth.
Zenawi (Ethiopia)	Yes	Partial	10% growth, largely from public investment, not heavily export-based.
Obiang (Eq. Guinea)	Yes	Yes	Explosive growth from oil, nearly all of GDP tied to exports.
Mubarak (Egypt)	Partial	Partial	Growth via gas, tourism, remittances, and modest export expansion.
Mohammed VI (Morocco)	Partial	Partial	Exports grew, but tourism and domestic consumption were also key.

Table A.1: Assessment of Growth and Export Strategy in Selected Autocratic Regimes. Note: I categorize "Fast Growth" and "Export-led" by Yes, Partial, and No.

A.2 Descriptive Performance

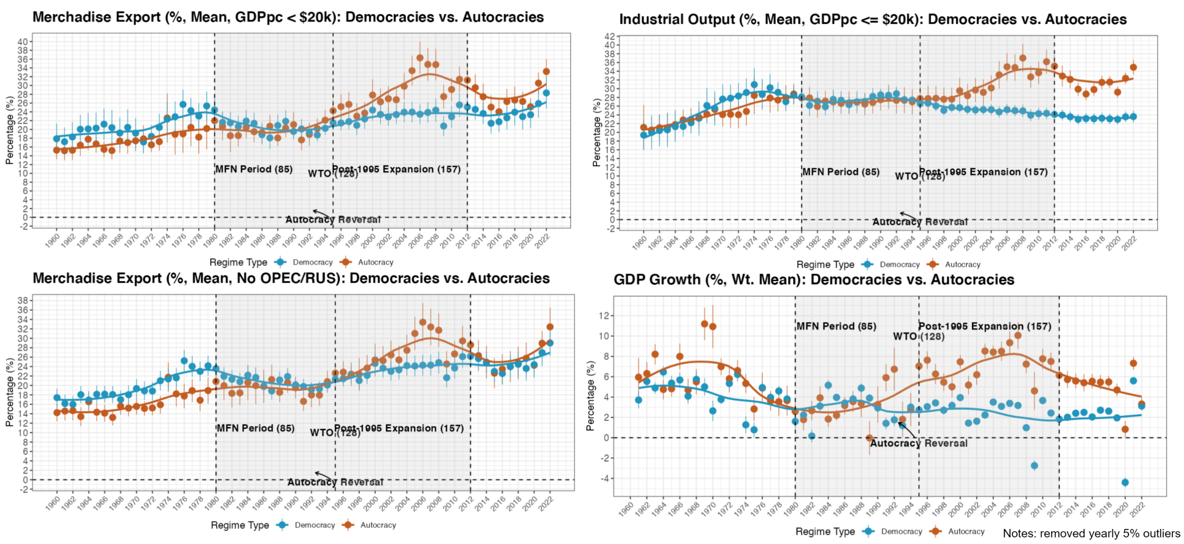
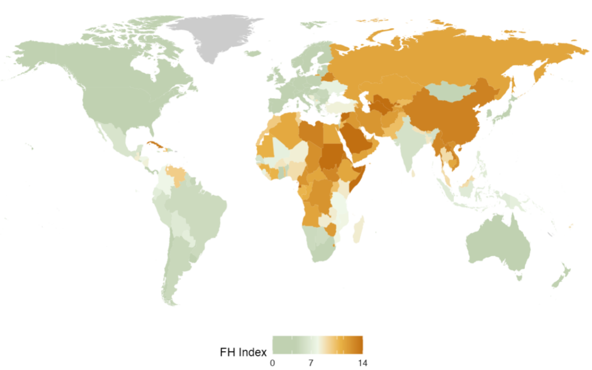
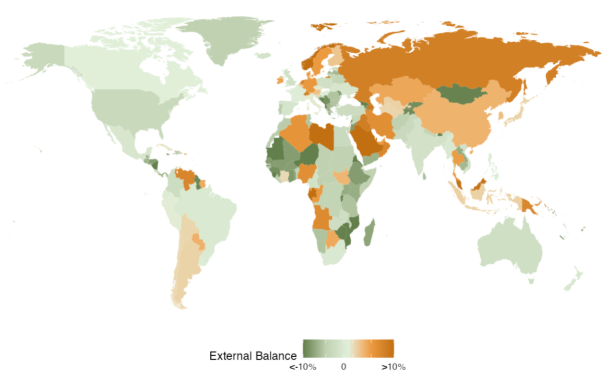


Figure A.4: Average Performance of Economic Indicators between Democracies and Autocracies (FH ≥ 11).



Freedom House Index



Average External Balance (% of GDP)

Figure A.5: Correlation between Regime Type and External Balance. *Note*: average external balance calculate the mean of current account balance and trade balance to include the information of both balances, since the two oftentimes do not overlap.

A.3 Panel Data of Regime Type Change

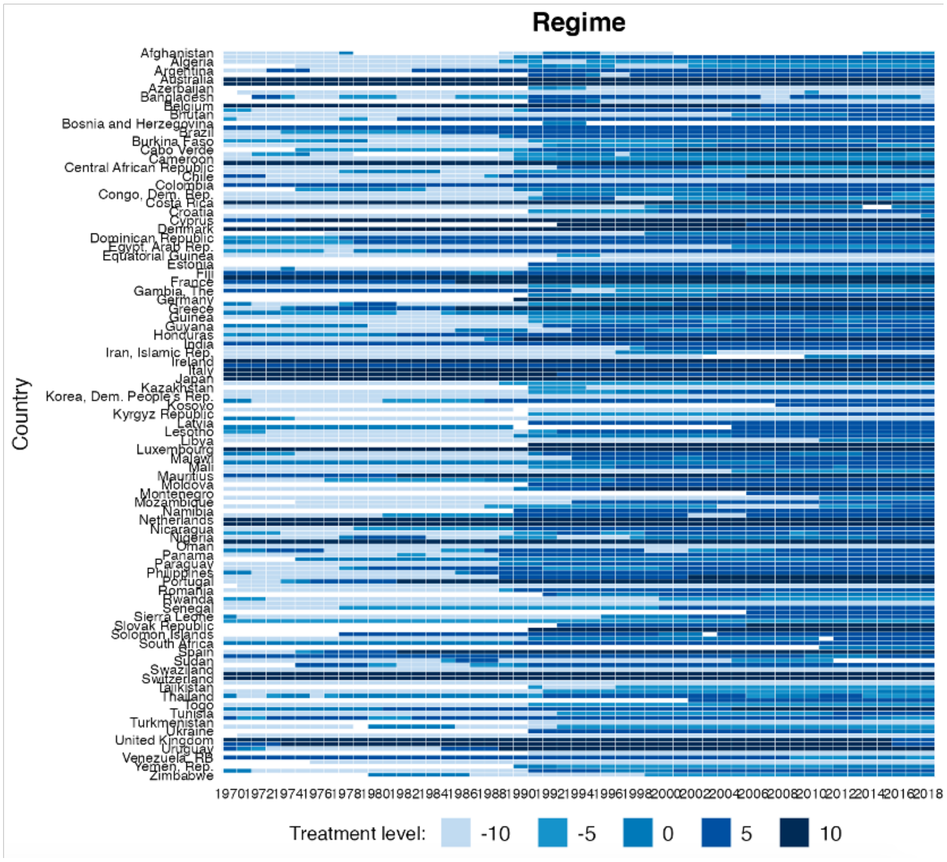


Figure A.6: Democratization (Polity Index)

A.4 KOF Globalization Index (Economic) Change

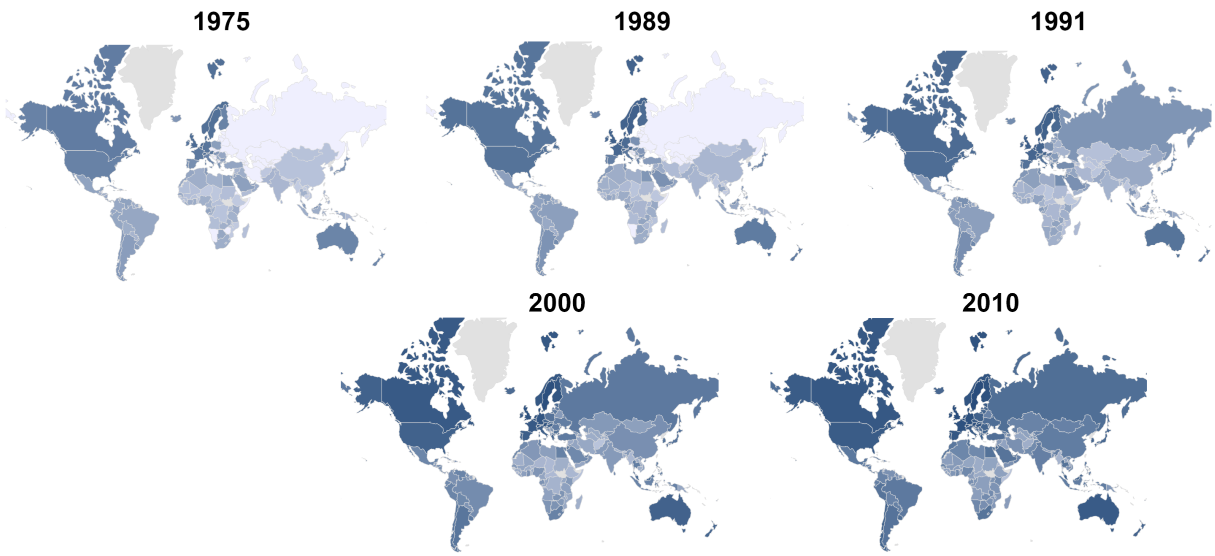


Figure A.7: KOF Globalization Index (Economic) Change. *Note*: The KOF economic index measures flows of trade, FDI and transfers, and trade and capital accounts restrictions.

A.5 WTO Membership

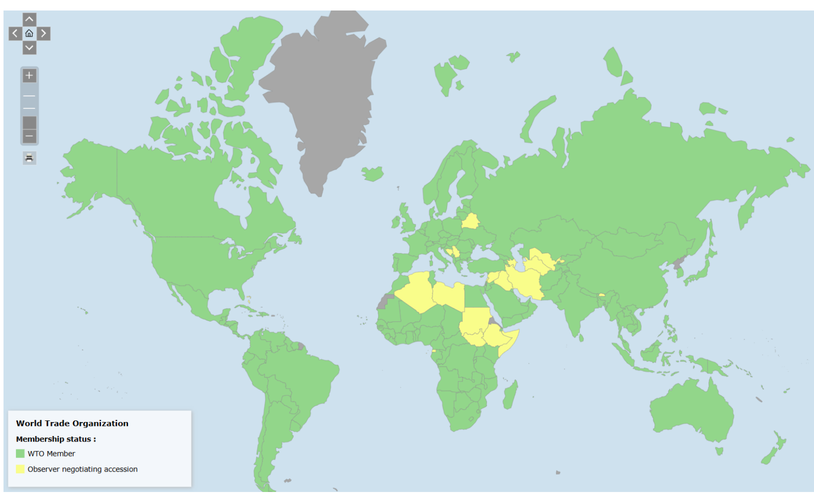


Figure A.8: The Map of WTO Members and Observers (source: WTO website)

B Theory Section

B.1 An Extended Eaton-Kortum Model

This section extends the classic Eaton-Kortum (E-K) model (Eaton and Kortum 2002) to illustrate the logic and hypotheses of this paper, since this model focuses on the determinants of bilateral trade flows. As in the paper, the trade flow from country i to j in E-K model is formally expressed as below:

$$X_{ij} = \frac{T_i(c_i\tau_{ij})^{-\theta}}{\sum_k T_k(c_k\tau_{kj})^{-\theta}} Y_j \tag{7}$$

where T_i is country i's state of technology (or absolute advantage). c_i is cost of production, and τ_{ij} is trade barrier between two countries i and j. θ denotes the heterogeneity of a country's efficiency in producing a continuum of goods drawn from a Fréchet distribution ($F_i(z) = e^{-T_i z^{-\theta}}$) – that is, the comparative advantage. X_{ij} measures the trade flow from i to j, while Y_j is the aggregate consumption of country j. Together, (1) implies that a country's exports are determined by its technology, production cost, and trade cost.

In (2) - (4), three new variables related to my paper are introduced: institution (I_i), autocratic advantage (A_i, proxied by regime type), and WTO membership (W_i). In a globalized economy, trade-related productive technology (T_i) of country i is largely affected by investments, assuming more significantly from involving the GVC. Therefore, T_i depends on institutional improvement (I_i) and autocratic advantage (A_i) to attract the GVC, conditional on crossing an institutional threshold (I^*) and possessing WTO membership ($W_i = 1$, to proxy trade engagement). Both I^* and $W_i = 1$ are needed for substantively attracting firms, domestic or multinational, to invest and innovate. In other words, if institution is too low or excluded from the global trade system, institutional improvement or autocratic advantage won't matter much. Autocratic advantages A_i, embodied in state-market synergy, magnify the effect of I_i. Technology function $t(A_i, I_i)$ can be thus formally written as:

$$t(A_i, I_i) = \begin{cases} \bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i} + \gamma \frac{I_i A_i}{1+\lambda I_i^2}\right), & \text{if } I_i \geq I^* \text{ and } W_i = 1, \\ \bar{T}_i, & \text{otherwise.} \end{cases} \tag{8}$$

where $\bar{T}_i$ is the baseline technology. $\frac{\beta A_i}{1+\lambda A_i}$ models diminishing returns to A_i, while $\lambda > 0$ controls how quickly diminishing returns to A_i sets in. When A_i is small: $\frac{\beta A_i}{1+\lambda A_i} \approx \beta A_i$, so the contribution of A_i grows almost linearly. When A_i is large: $\frac{\beta A_i}{1+\lambda A_i} \longrightarrow \frac{\beta}{\lambda}$, so the marginal effect of A_i diminishes significantly and possibly goes negative when A_i too high. Similarly, $\frac{I_i A_i}{1+\lambda I_i^2}$ models diminishing returns to I_i. The economic rationale behind is that excessive centralization (high A_i) may create inefficiencies or governance rigidities, while saturated institutions (high I_i) may over-complicate decision-making, reducing efficiency. A moderately autocratic regime (e.g., with some centralized control) may gain substantial benefits, but extreme autocracy may lead to inefficiencies (e.g., power abuse).

Although I_i and A_i exhibit diminishing returns, they may still interact to amplify technology – i.e., the interaction term $\gamma \frac{I_i A_i}{1+\lambda_I I_i^2}$. γ captures the interaction coefficient between I_i and A_i on (T_i). Moderate levels of I_i and A_i together create the largest gains in technology because they complement each other. For example, moderate autocratic regimes (e.g., single-party regimes) may gain disproportionately when combined with moderate to high institutions. $\frac{I_i A_i}{1+\lambda_I I_i^2}$ also captures that when I_i is high, the effect of A_i diminishes (the denominator is dominated by I_i^2), because even autocratic regimes are now more hands-tied, if not completely disabled.

As argued, production cost c_i decreases with autocratic advantage A_i, due to reasons such as labor rights suppression and disproportionate state support for industries and infrastructure. c_i is formally expressed as:

$$c(A_i) = \begin{cases} \bar{c}_i \cdot \exp(-\delta A_i), & \text{if } I_i \geq I^* \text{ and } W_i = 1, \\ \bar{c}_i, & \text{otherwise.} \end{cases} \tag{9}$$

where $\bar{c}_i$ is the baseline production cost and δ_A captures the cost-reducing effect of autocratic advantage. The term $\exp(-\delta_A A_i)$ represents an exponential decay. As A_i increases, $\exp(-\delta_A A_i)$ becomes smaller, but the rate of decrease slows down because the exponential decay flattens over A_i. The intuition is that there's a limit to how much A_i can reduce costs, as well as marginal diminishing returns. The effective productive cost reduction is also assumed to be conditioned by crossing certain I^* and $W_i = 1$, otherwise, A_i may have limited effect.

Trade cost τ_{ij} decreases with WTO membership W_i, formally as:

$$\tau(W_i) = \bar{\tau}_{ij} \cdot (1 - \varphi W_i) \tag{10}$$

where τ_{ij} is baseline trade cost, and $\lambda_W > 0$ indicates reduction in trade costs due to WTO membership (or trade engagement). Therefore, (1) takes on the form of:

$$X_{ij} = \frac{t(A_i, I_i)\{c(A_i)\tau(W_i)\}^{-\theta}}{\sum_k t(A_k, I_k)\{c(A_k)\tau(W_k)\}^{-\theta}} Y_j \tag{11}$$

Plug (2)-(4) into (5), the full trade flow formula becomes:

$$X_{ij} = \frac{\left[\bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i} + \gamma \frac{I_i A_i}{1+\lambda I_i^2}\right)\right] \cdot \left\{[\bar{c}_i \cdot \exp(-\delta A_i)] \cdot \left[\bar{\tau}_{ij} \cdot (1 - \varphi W_i)\right]\right\}^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j, \quad \text{if } I_i \geq I^* \text{ and } W_i = 1. \tag{12}$$

From (5), it shows that if $W_i = 0$, meaning a country i is not engaged in the global trade system, trade flow from i to j is simplified to the original baseline form (6), which means neither institutional improvement nor increased autocratic advantages (proxied by more autocratic regime type) will significantly improve trade flows.

$$X_{ij} = \frac{\bar{T}_i (\bar{c}_i \bar{\tau}_{ij})^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j \tag{13}$$

When W_i changes from 0 to 1, trade cost decreases, so that trade flow from X_{ij} increases. An increase in autocratic advantages A_i leads to an increase in productive technology:

$$\bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i} + \gamma \frac{I_i A_i}{1+\lambda I_i^2 + \varepsilon A_i}\right)$$

and a decrease in productive cost $\bar{c}_i \cdot \exp(-\delta A_i)$. Therefore, exports X_{ij} increases. This is consistent with *H1.1*.

Note that in order for productive technology to increase and productive cost to decrease, I_i has to cross certain thresholds (i.e., $I_i > I^*$), and it cannot be too high. This comes from conditions in (2) and (3) and is consistent with *H1.2*. To combine all, when W_i changes from 0 to 1 and $I_i > I^*$, (5) minus (6) becomes:

$$\Delta X_{ij} = \frac{\left[\bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i} + \gamma \frac{I_i A_i}{1+\lambda I_i^2}\right)\right] \cdot \left\{[\bar{c}_i \cdot \exp(-\delta A_i)] \cdot \left[\bar{\tau}_{ij} \cdot (1-\varphi)\right]\right\}^{-\theta} - \bar{T}_i(\bar{c}_i \bar{\tau}_{ij})^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j \tag{14}$$

(7) formally denotes that after passing institutional thresholds I^*, an increase in A_i leads to more export increase for the same WTO accession. Put differently, autocratic WTO-joiners are expected to experience more export increase than their democratic counterparts.

Similarly, when $W_i = 0$, an increase in institution will not increase $\bar{T}_i$ or decrease $\bar{c}_i$, thus not increasing X_{ij}, as specified by (2) and (3). However, when $W_i = 1$, for a ΔI increase in institutional level while $\bar{c}_i \cdot \exp(-\delta A_i)$ and $\bar{\tau}_{ij} \cdot (1 - \varphi W_i)$ keep unchanged, change in X_{ij} is expressed and simplified as:

$$\Delta X_{ij} = \frac{\left[\bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i}\right) \exp\left(\frac{I_i + \Delta I}{1+\lambda (I_i+\Delta I)^2} - \frac{I_i}{1+\lambda I_i^2}\right)\right] \cdot \left\{[\bar{c}_i \cdot \exp(-\delta A_i)] \cdot \left[\bar{\tau}_{ij} \cdot (1-\varphi)\right]\right\}^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j \tag{15}$$

(8) also implies an increase in A_i leads to more export increase for the same institutional improvement, given crossing institutional thresholds and trade engagement which is consistent with *H2.1* and *H2.2*. In other words, more autocratic states engaged by the global trade regime should expect more gains from institutional improvement in a globalized economy. Again, when A_i may

be subject to possible decreasing marginal returns.

Last, since X_{ij} denotes the absolute level of trade flow from i to j, (5) also implies that more autocratic states (larger A_i) are expected to "inflate" trade flows more conditional on other factors such as trade engagement and institutional levels. However, this is also subject to the conditions in (2) and (3). For example, for states that all have $W_i = 1$, A_i's effect may diminish when I_i doesn't cross thresholds I^* or is too high (*H3.1*). For states that all have similarly moderate institutional levels I_i, A_i's effect may diminish when $W_i = 0$ (*H3.2*).

B.2 Predictive Patterns

Prediction on Exports

$$X_{ijk} = \frac{s_{ik} Y_i Y_j}{(p_{ik})^{\sigma} \overline{y}_{ik}} \left[T_{ijk}\left(z_i, z_j\right) / P_j^k \right]^{1-\sigma} [\theta_{ik} exp(z_i)]^{\sigma - 1}$$

Example of Gravity Model Incorporating Product Quality, Yu(2010)

First, similar to Yu (2010), by employing gravity model commonly used in economics and political science (Anderson and van Wincoop 2003) controlling for a standard list of dyad-level covariates, I find that prior to 1990, being more democratic is associated with higher exports (see Table B.2). Post-1990, however, being more autocratic is associated with a positive or zero effect compared to being more democratic is.[51] The models include cross-sectional, within-exporter, interaction with exporter's logged GDP (whether the coefficient differs for larger countries), and weighted least squares (when larger countries are assigned larger weights). Using the interaction model, for example, by plugging in Iran's GDP in 2005 (the logged form = 20), the effect of Polity is negative.

[51] For post-1990, I look at all dyads with exporter being within the WTO, since many autocracies joined the WTO after 1990 and being inside the WTO is what I am interested in. In contrast, the pre-1990 model checks both inside and outside of the WTO since most autocracies were excluded. However, the result barely changes if WTO only.

	Pre-1990	Post-1990 (within WTO)			
	OLS	OLS	OLS (Within)	OLS (Interaction)	WLS (by GDP)
Polity_i	0.022***	0.003	−0.016***	0.065***	−0.041***
	(0.001)	(0.003)	(0.004)	(0.014)	(0.001)
Polity_i x *GDP_i*				−0.004***	
				(0.001)	
Polity_j	0.003*	0.003**	0.005***	0.003**	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
GDP_i	1.583***	−1.746***	0.298***	−1.698***	−1.137***
	(0.048)	(0.139)	(0.054)	(0.141)	(0.107)
GDP_j	2.058***	0.650***	0.521***	0.644***	0.065
	(0.143)	(0.080)	(0.090)	(0.079)	(0.182)
GDPPC_i	−0.536***	3.051***	0.070	3.023***	2.154***
	(0.043)	(0.129)	(0.063)	(0.129)	(0.116)
GDPPC_j	−1.011***	0.387***	0.526***	0.392***	0.967***
	(0.137)	(0.083)	(0.092)	(0.082)	(0.185)
RTA	0.204***	0.282***	0.286***	0.278***	0.202***
	(0.051)	(0.041)	(0.039)	(0.041)	(0.031)
Custom Union	0.819***	0.590***	0.662***	0.591***	−0.335***
	(0.111)	(0.031)	(0.032)	(0.030)	(0.030)
Common Colnizer post-45	0.775***	0.998***	0.875***	0.996***	0.533***
	(0.027)	(0.022)	(0.017)	(0.022)	(0.047)
Colonial Dep. post-45	1.724***	1.034***	1.260***	1.048***	0.982***
	(0.044)	(0.061)	(0.048)	(0.060)	(0.050)
Population_i	−0.757***	2.984***	−0.112	2.955***	2.190***
	(0.041)	(0.124)	(0.093)	(0.124)	(0.109)
Population_j	−1.195***	0.445***	0.596***	0.450***	0.938***
	(0.139)	(0.072)	(0.082)	(0.071)	(0.180)
Distance	−0.785***	−1.106***	−1.244***	−1.109***	−0.960***
	(0.025)	(0.011)	(0.015)	(0.011)	(0.013)
Common Language	0.294***	0.607***	0.685***	0.612***	0.331***
	(0.023)	(0.026)	(0.030)	(0.026)	(0.020)
Common Religion	−0.059*	−0.008	0.266***	−0.010	−0.025
	(0.030)	(0.032)	(0.033)	(0.032)	(0.028)
Border	0.462***	0.806***	0.529***	0.802***	0.073
	(0.017)	(0.023)	(0.023)	(0.022)	(0.044)
Num.Obs.	194 716	313 566	313 566	313 566	313 580
R2 Adj.	0.629	0.709	0.759	0.709	0.801
FE	year	year	year/exporter	year	year

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.2: Regime Type and Exports

B.3 Selection of Institutional Thresholds

In 2000, the bottom 20 percentile threshold is 0.45 for PR protection and 0.2 for rule of law, respectively. I combine the institutional levels at the bottom 20 percentile among developing countries in 2000 and real cases (e.g., China's PR protection is around 0.35), so the thresholds are roughly 0.2 for rule of law and 0.35 for PR protection. Both values have to be reached. However, special cases remain. First, I slightly prioritize PR protection especially for resource-rich countries, for it is more attractive to the GVC than rule of law – as long as global investor's property rights are protected, global firms may more rely on within-GVC contract enforcement. For example, Cameroon and Chad, two resource-rich African countries have high PR protection (0.8 and 0.78) but low rule of law (below 0.1), for which I classify them as reformed. Azerbaijan (0.61, .03) and Equatorial Guinea (0.45, 0.06) are two other cases. Second, I factor in expectation. Venezuela's values for two indicators were 0.58 and 0.05 in 2010. Yet, Venezuela has experienced rapid institutional deterioration since 1997 before Hugo Chávez was elected completely reversing course when the two indicators were as high as 0.9 and 0.55, generating greatly adverse expectations for investors. Thus, Venezuela is listed as non-reformer. Yemen is another example: from the Arab Spring in 2011 to Houthi's takeover in 2015, its institutions experienced rapid deterioration.

B.4 Share of World Exports for Major Autocracies

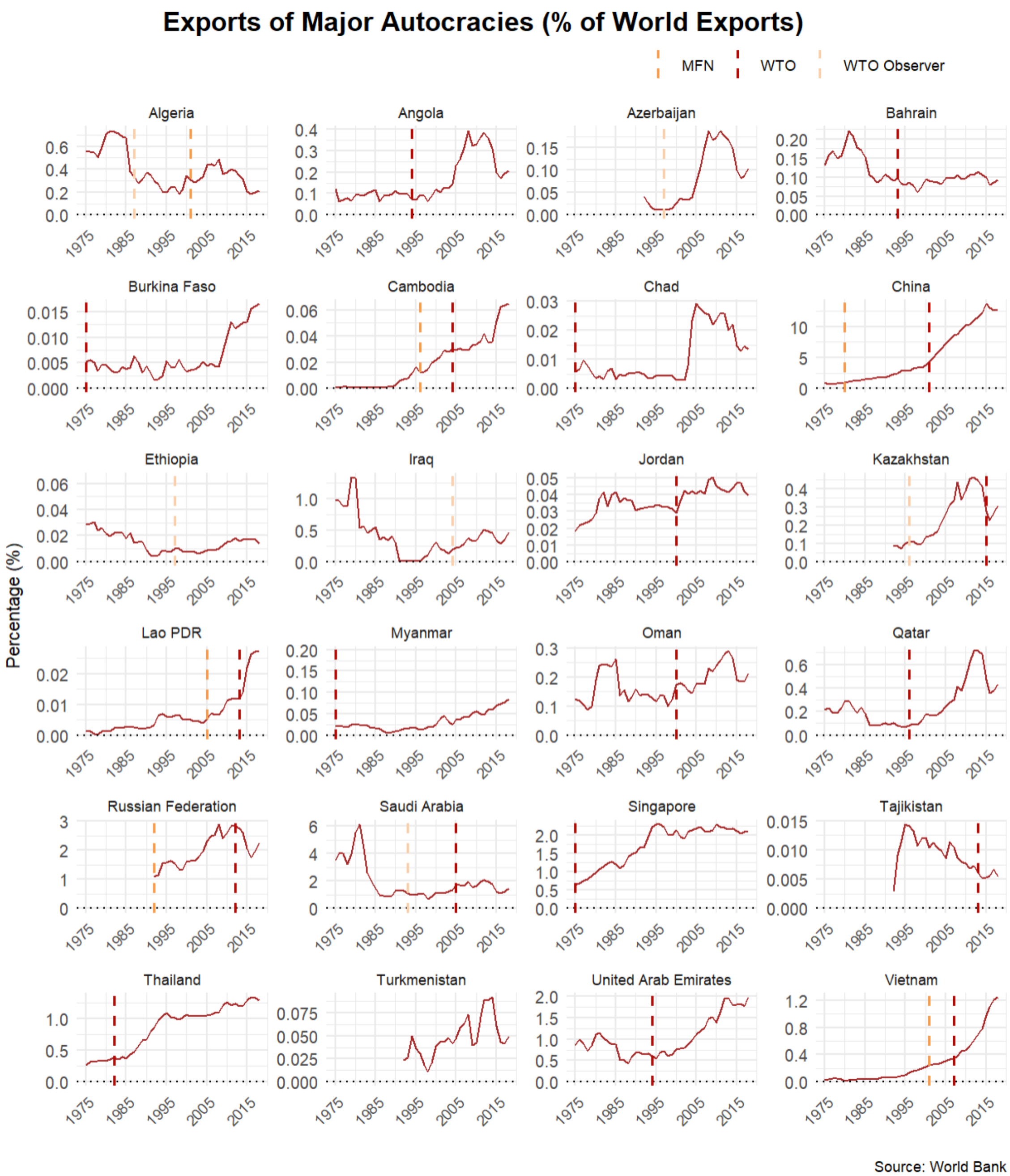
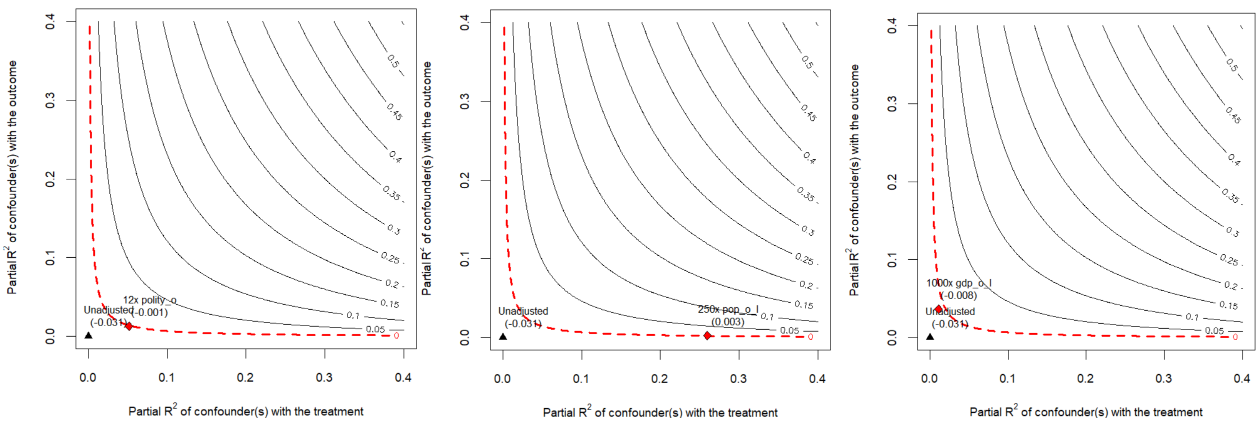


Table B.3: Share of World Exports for Major Autocracies. *Note*: For illustration purpose, vertical dashed lines begin in 1975 if MFN/WTO/Observer in effect earlier. Most autocracies' global share in exports show an increasing trend.

C Main Empirical Tests

C.1 Sensitivity Analysis



PR Protection | Population(log) | GDP(log)

Figure C.9: Improved Covariate Balance via CBPS Weighting, Post-1990. *Note*: overall, all covariates' balance improve significantly. The green line natural resource intensity is slightly not balanced. However, it may not be an concern as it does not significantly affect WTO accession theoretically.

C.2 WTO Effect

	Exports (FE)		Exports (FE)		Exports (FE)		Exports (FE)		Exports (CRE)	
	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990
$One\ WTO$	0.304***	−0.131**								
	(0.052)	(0.061)								
$One\ WTO \times Polity_i$	**-0.003**	**-0.025*** **								
	(0.005)	(0.008)								
$Both\ WTO$	0.615***	0.143	0.314***	0.273***					0.309***	0.209***
	(0.084)	(0.092)	(0.046)	(0.047)					(0.015)	(0.012)
$Both\ WTO \times Polity_i$	**0.007**	**-0.041*** **	**0.011*** **	**-0.033*** **					**0.012** **	**-0.038*** **
	(0.006)	(0.010)	(0.004)	(0.006)					(0.001)	(0.002)
$WTO_i\ Only$			0.306***	−0.037					0.304***	0.099***
			(0.052)	(0.065)					(0.017)	(0.017)
$WTO_i\ Only \times Polity_i$			**0.005**	**-0.037*** **					**0.004** **	**-0.038*** **
			(0.005)	(0.007)					(0.001)	(0.002)
WTO_i					0.307***	0.210***	0.240***	−0.022		
					(0.042)	(0.045)	(0.049)	(0.059)		
$WTO\_i \times Polity_i$					**0.009** **	**-0.033*** **	**-0.001**	**-0.003**		
					(0.008)	(0.004)	(0.006)	(0.008)		
$WTO_i\ (3y\ lag)$							0.105**	0.365***		
							(0.042)	(0.051)		
$WTO_i\ (3y\ lag) \times Polity_i$							**0.012** **	**-0.032*** **		
							(0.005)	(0.007)		
GDP_i	0.035	0.216**	−0.076	0.211**	−0.107	0.202**	−0.061	0.127	−0.213***	0.416***
	(0.096)	(0.091)	(0.091)	(0.091)	(0.091)	(0.090)	(0.093)	(0.095)	(0.081)	(0.115)
$GDPPC_i$	0.502***	0.278***	0.620***	0.285***	0.654***	0.288***	0.600***	0.282***	0.758***	0.070
	(0.088)	(0.093)	(0.085)	(0.093)	(0.084)	(0.092)	(0.086)	(0.098)	(0.079)	(0.115)
PTA	0.144***	0.191***	0.225***	0.192***	0.226***	0.193***	0.224***	0.074**	0.170***	0.191***
	(0.030)	(0.030)	(0.032)	(0.030)	(0.032)	(0.030)	(0.032)	(0.031)	(0.012)	(0.013)
RTA	0.165***	0.024	0.121***	0.024	0.127***	0.020	0.111**	0.074**	0.133***	0.005
	(0.042)	(0.031)	(0.045)	(0.031)	(0.045)	(0.031)	(0.045)	(0.031)	(0.021)	(0.015)
$Customs\ Union$	0.312**	0.088	0.233	0.096	0.242*	0.099	0.233	0.105	0.343***	0.134***
	(0.144)	(0.083)	(0.144)	(0.083)	(0.144)	(0.083)	(0.142)	(0.084)	(0.049)	(0.034)
$Colonial\ Dependency$	0.600***	1.434***	0.626***	1.440***	0.622***	1.445***	0.618***	1.345***	0.613***	1.217***
	(0.091)	(0.216)	(0.085)	(0.217)	(0.086)	(0.216)	(0.089)	(0.227)	(0.118)	(0.406)
$Polity_i$	0.013**		0.009**		0.009**		0.007*			
	(0.005)		(0.004)		(0.004)		(0.004)			
$Com.\ Col.\ Post45$									0.469***	0.733***
									(0.056)	(0.042)

$Col. Dep. Post45$									2.207***	1.736***
									(0.137)	(0.125)
$Distance$									−1.063***	−1.380***
									(0.021)	(0.016)
$Common Language$									0.196***	0.700***
									(0.045)	(0.036)
$Common Religion$									0.075	0.324***
									(0.063)	(0.048)
$Bordered$									0.658***	0.870***
									(0.091)	(0.080)
Exporter Means									✓	✓
Dyad Means									✓	✓
Exporter FE	✓	✓	✓	✓	✓	✓	✓	✓	RE	RE
Importer FE	✓	✓	✓	✓	✓	✓	✓	✓		
Dyad FE	✓	✓	✓	✓	✓	✓	✓	✓	RE	RE
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num.Obs.	210 809	538 470	210 809	538 470	210 809	538 470	208 242	502 944	210 049	521 997
R2 Adj.	0.858	0.886	0.868	0.886	0.868	0.886	0.869	0.892	0.874	0.882
BIC	805 515.9	2 258 339.3	833 195.6	2 258 125.3	833 258.2	2 258 320.9	824 099.5	2 103 706.2	691 636.8	1 901 073.2

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.4: The Effects of Joining the WTO

	Exports (FE Model)		Exports (CRE Model)	
	PR Protection	RoL	PR Protection	RoL
$WTO\_i : polity2000$	0.033	0.165***	−0.068***	0.122***
	(0.040)	(0.036)	(0.015)	(0.019)
$WTO\_i : polity2000 : inst\_prpty\_low$	−0.057		0.044**	
	(0.046)		(0.017)	
$WTO\_i : polity2000 : inst\_prpty\_mid$	−0.092**		0.009	
	(0.041)		(0.015)	
$WTO_i : polity2000 : inst\_rule\_low$		−0.083**		−0.033*
		(0.040)		(0.020)
$WTO_i : polity2000 : inst\_rule\_mid$		−0.226***		−0.183***
		(0.037)		(0.019)
WTO_i	−0.365	−1.265***	0.270**	−1.020***
	(0.285)	(0.307)	(0.112)	(0.152)
$Polity_i$	−0.025***	−0.025***	−0.024***	−0.024***
	(0.004)	(0.004)	(0.001)	(0.001)
$Polity_j$	−0.001	−0.001	0.004***	0.003***
	(0.003)	(0.003)	(0.001)	(0.001)
WTO_j	−0.055	−0.059	−0.110***	−0.107***
	(0.064)	(0.063)	(0.030)	(0.030)
$Both WTO$	0.217***	0.220***	0.268***	0.266***
	(0.065)	(0.064)	(0.031)	(0.031)
GDP_i	1.770	1.603	−0.026	−0.237
	(16.391)	(16.287)	(6.526)	(6.519)
GDP_j	1.304***	1.298***	1.393***	1.386***
	(0.243)	(0.244)	(0.171)	(0.170)
$GDPPC_i$	−1.339	−1.192	0.455	0.651
	(16.391)	(16.288)	(6.525)	(6.519)
$GDPPC_j$	−0.568**	−0.562**	−0.513***	−0.507***
	(0.242)	(0.244)	(0.171)	(0.170)
PTA	0.117***	0.111***	0.146***	0.142***
	(0.038)	(0.038)	(0.019)	(0.019)
RTA	0.067*	0.061	0.088***	0.083***
	(0.040)	(0.039)	(0.021)	(0.021)
$Customs Union$	−0.008	−0.019	0.067	0.058
	(0.097)	(0.097)	(0.044)	(0.044)

$Population_i$	−1.673	−1.509	0.226	0.419
	(16.390)	(16.286)	(6.526)	(6.519)
$Population_j$	−0.177	−0.192	−0.325*	−0.321*
	(0.250)	(0.251)	(0.171)	(0.170)
$WTO_i : inst\_prpty\_low$	0.086		−0.612***	
	(0.302)		(0.124)	
$WTO\_i : inst\_prpty\_mid$	0.377		−0.308***	
	(0.288)		(0.111)	
$WTO\_i : inst\_rule\_low$		0.921***		0.630***
		(0.325)		(0.152)
$WTO\_i : inst\_rule\_mid$		1.325***		1.041***
		(0.302)		(0.150)
$polity2000$			0.152	−0.058
			(0.141)	(0.097)
$inst\_prpty\_catlow$			−0.266	
			(1.192)	
$inst\_prpty\_catmid$			0.165	
			(1.156)	
$Common\ Colonizer$			0.789***	0.785***
			(0.057)	(0.057)
$Colonial\ Dep.$			2.072***	2.077***
			(0.191)	(0.191)
$Distance$			−1.437***	−1.440***
			(0.023)	(0.023)
$Common\ Language$			0.687***	0.687***
			(0.051)	(0.051)
$Common\ Religion$			0.271***	0.273***
			(0.068)	(0.068)
$Bordered$			0.911***	0.912***
			(0.106)	(0.106)
$polity2000 : inst\_prpty\_catlow$			−0.043	
			(0.160)	
$polity2000 : inst\_prpty\_catmid$			−0.099	
			(0.148)	
$inst\_rule\_catlow$				−1.207
				(0.974)
$inst\_rule\_catmid$				−0.809
				(0.822)
$polity2000 : inst\_rule\_catlow$				0.066
				(0.177)
$polity2000 : inst\_rule\_catmid$				0.167
				(0.104)
Exporter Means			✓	✓
Dyad Means			✓	✓
Exporter FE	✓	✓	RE	RE
Importer FE	✓	✓		
Dyad FE	✓	✓	RE	RE
Year FE	✓	✓	✓	✓
Num.Obs.	284 345	284 345	275 914	275 914
R2	0.866	0.866		
R2 Adj.	0.858	0.858		
R2 Marg.			0.420	0.400
R2 Cond.			0.860	0.860
BIC	1 201 319.8	1 200 671.1	1 034 876.7	1 034 368.6

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.5: The Effects of Joining the WTO by Institutional Range.

	DV: Current Account Balance (%)						
	FE	FE	RE	RE/No OPEC	RE/Developing	RE/WTO Post-90	80-95
Polity	−0.212***	−0.123***	−0.146***	−0.135***	−0.139***	−0.225*	0.095*
	(0.039)	(0.043)	(0.049)	(0.049)	(0.053)	(0.119)	(0.052)
GDP (log)		1.744***	1.430***	1.427***	1.440***	1.032	0.433
		(0.129)	(0.265)	(0.271)	(0.296)	(0.707)	(0.414)
GDPPC (log)		0.787***	−0.148	−0.624	−0.385	1.229	−0.121
		(0.251)	(0.399)	(0.415)	(0.449)	(0.916)	(0.697)
GDP Growth		−0.182**	−0.118***	−0.146***	−0.127***	−0.034	−0.074
		(0.093)	(0.031)	(0.032)	(0.035)	(0.063)	(0.046)
Net Borrowing(%)		0.657***	0.475***	0.453***	0.519***	0.657***	−0.014
		(0.061)	(0.027)	(0.031)	(0.032)	(0.053)	(0.072)
Foreign Asset(%)		0.418*	0.068	0.060	0.053	−0.065	0.724**
		(0.231)	(0.119)	(0.114)	(0.128)	(0.158)	(0.301)
KA Open		−0.264*	−0.342**	−0.230	−0.382**	−0.370	−0.382
		(0.140)	(0.172)	(0.171)	(0.185)	(0.390)	(0.287)
Δ*Private Credit* (%)		−0.195***	−0.139***	−0.138***	−0.178***	−0.238***	−0.127***
		(0.038)	(0.015)	(0.014)	(0.021)	(0.041)	(0.036)
Δ*Term of Trade*		0.064**	0.085***	0.061***	0.096***	0.082***	0.031**
		(0.026)	(0.011)	(0.012)	(0.012)	(0.019)	(0.015)
Population (*under* 14, %)		15.831***	23.902***	19.920***	25.587***	55.684***	13.231
		(4.223)	(5.444)	(5.558)	(6.330)	(11.228)	(11.693)
Population (*over* 65, %)		5.094	41.690***	41.891***	49.794***	98.099***	42.435
		(6.103)	(8.405)	(8.305)	(11.467)	(21.183)	(34.278)
Trade Openness		0.041***	0.041***	0.043***	0.028***	−0.013	−0.004
		(0.005)	(0.006)	(0.006)	(0.008)	(0.015)	(0.018)
Year		−0.316***	−0.278***	0.024	−0.039	−0.118	0.299***
		(0.062)	(0.082)	(0.165)	(0.048)	(0.112)	(0.070)
Year FE	✓	✓	✓	✓	✓	✓	✓
Country RE			✓	✓	✓	✓	✓
Num.Obs.	2704	1499	1499	1698	1469	540	290
R2 Marg.			0.422	0.384	0.322	0.420	0.252
R2 Cond.			0.759	0.757	0.691	0.755	0.605
BIC	43 454.6	20 230.6					

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.6: Regime Type's Effect on Current Account Balance.

	DV: Trade Balance (%)						
	FE	FE	RE	RE/No OPEC	RE/Developing	RE/WTO Post-90	80-95
Polity	−0.410***	−0.480***	−0.241***	−0.179***	−0.227***	−0.268**	0.140**
	(0.050)	(0.069)	(0.056)	(0.055)	(0.064)	(0.124)	(0.057)
GDP (log)		2.594***	1.692***	1.896***	2.189***	0.309	0.411
		(0.183)	(0.453)	(0.442)	(0.534)	(1.587)	(0.845)
GDPPC (log)		4.872***	3.032***	2.002***	3.007***	7.007***	−2.969***
		(0.411)	(0.592)	(0.595)	(0.698)	(1.703)	(1.142)
GDP Growth		0.054	−0.070**	−0.176***	−0.092**	0.074	−0.048
		(0.102)	(0.034)	(0.035)	(0.041)	(0.067)	(0.043)
Net Borrowing (%)		0.704***	0.467***	0.376***	0.524***	0.719***	−0.082
		(0.075)	(0.031)	(0.035)	(0.037)	(0.056)	(0.073)

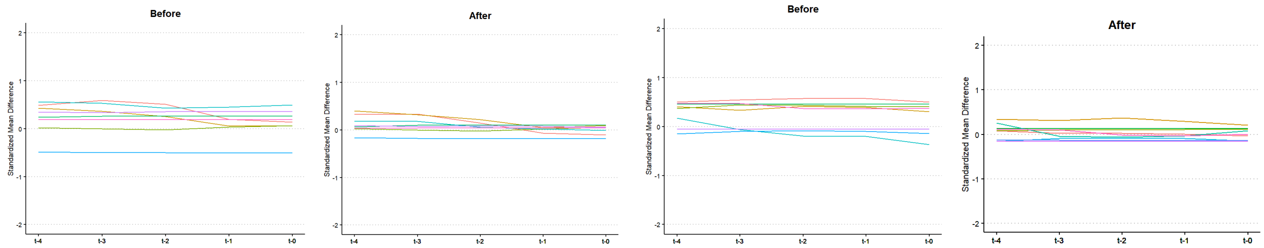
Foreign Asset (%)		−0.589	0.137	0.139	0.107	0.063	0.044
		(0.522)	(0.137)	(0.127)	(0.155)	(0.177)	(0.313)
KA Open		−0.746***	−0.008	0.189	0.002	0.427	−0.235
		(0.253)	(0.214)	(0.205)	(0.246)	(0.506)	(0.295)
Δ *Private Credit* (%)		−0.234***	−0.194***	−0.185***	−0.230***	−0.303***	−0.145***
		(0.055)	(0.017)	(0.016)	(0.025)	(0.046)	(0.035)
Δ *Term of Trade*		0.105***	0.094***	0.061***	0.105***	0.084***	0.032**
		(0.038)	(0.012)	(0.013)	(0.014)	(0.020)	(0.014)
Population (*under* 14, %)		32.852***	25.700***	22.400***	35.155***	29.318**	−50.296***
		(5.722)	(6.973)	(6.947)	(8.605)	(14.287)	(14.655)
Population (*over* 65, %)		−7.151	15.419	18.005*	44.023***	119.972***	14.179
		(7.680)	(10.876)	(10.555)	(16.531)	(30.229)	(56.648)
Trade Openness		0.053***	0.042***	0.041***	0.023**	−0.051***	−0.047**
		(0.007)	(0.008)	(0.007)	(0.010)	(0.018)	(0.021)
Year		−0.950***	−1.251***	−0.136	−0.236***	−0.537***	0.119
		(0.113)	−0.261	(0.185)	(0.061)	(0.138)	(0.073)
Year FE	✓	✓	✓	✓	✓	✓	✓
Country RE			✓	✓	✓	✓	✓
Num.Obs.	2829	1560	1560	1463	1243	551	294
R2 Marg.			0.447	0.436	0.347	0.341	0.252
R2 Cond.			0.882	0.880	0.848	0.938	0.900
BIC	46 158.5	22 594.1					

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.7: Regime Type's Effect on Trade Balance.

Since property-rights protection and rule of law have quite different distributions across autocratic WTO-joiners, I make sure both low and high institutional ranges contain at least some autocracies that joined the WTO during 1990-2020. The separation looks like $\{0, 0.3, 0.7, 1\}$. For each range, I compare autocracies to all democracies that joined during the same period to keep the control group the same and I dichotomize polity into a democracy dummy so that the interaction effect (WTO x polity) doesn't reflect within-democracy variation.

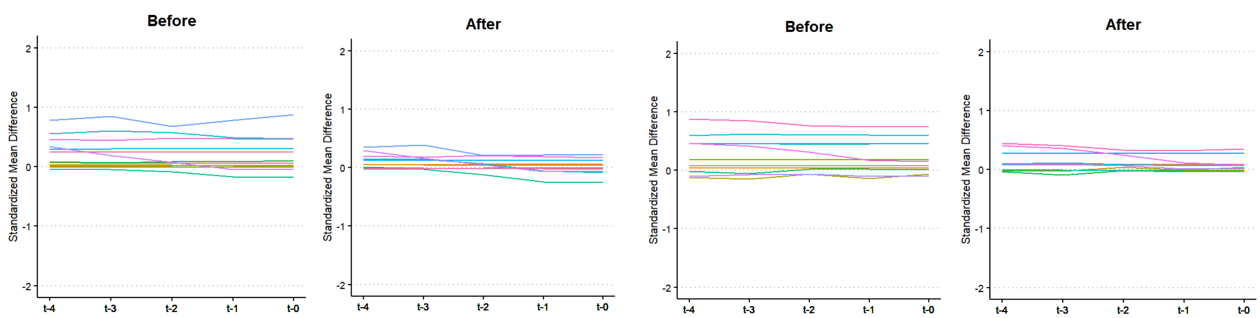
PanelMatch (country-years)



Democracy Democracy Autocracy Autocracy

Figure C.10: Improved Covariate Balance via CBPS Weighting, Post-1990. *Note*: overall, all covariates' balance improve significantly. The green line natural resource intensity is slightly not balanced. However, it may not be an concern as it does not significantly affect WTO accession theoretically.

PanelMatch (dyad-years)

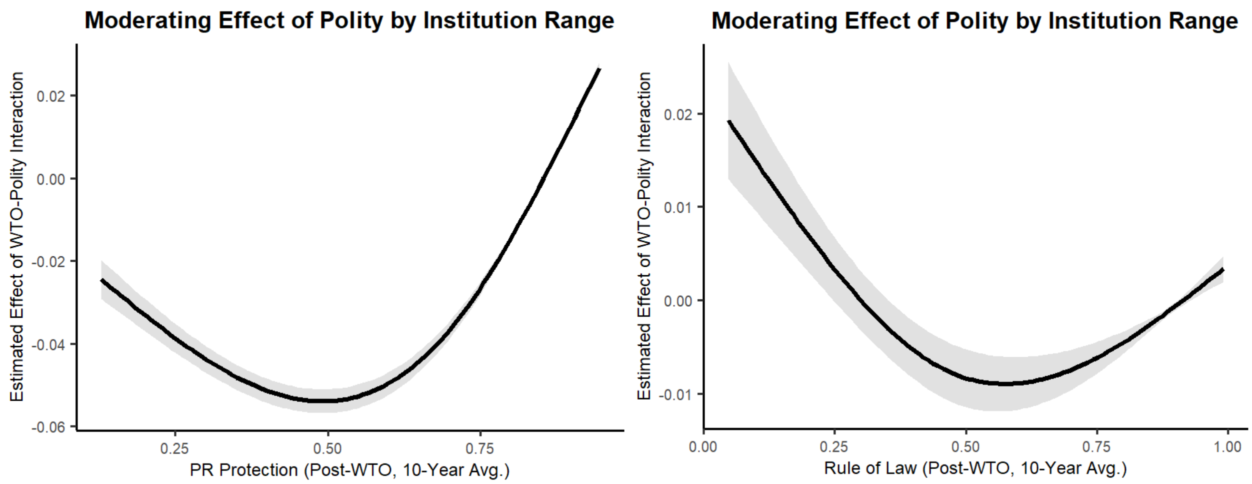


Democracy Democracy Autocracy Autocracy

Figure C.11: Improved Covariate Balance via CBPS Weighting, Post-1990. *Note*: overall, all covariates' balance improve significantly.

C.3 Moderated WTO Effect across Institutional Range

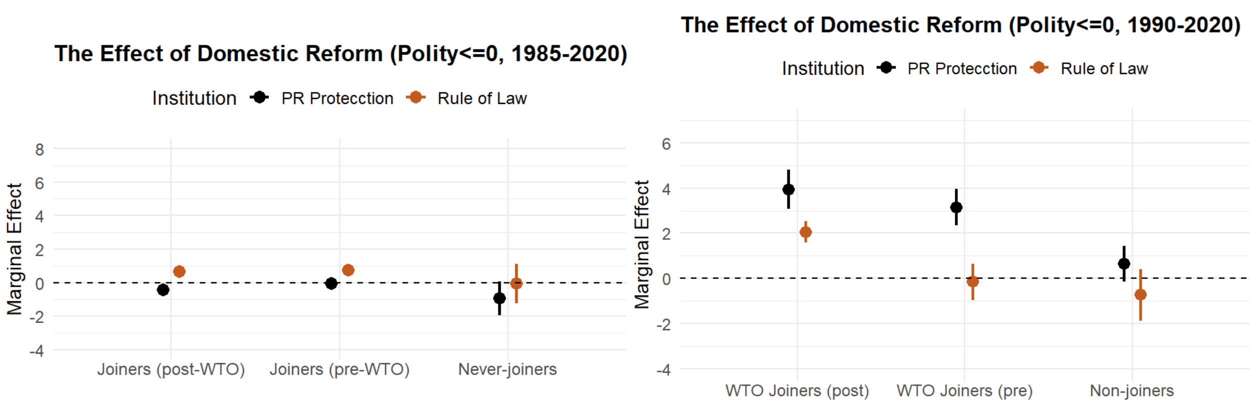
General Additive Model (GAM)



(a) (b)

Figure C.12: The Moderated Effect of Polity across Institutional Ranges (GAM).

C.4 Domestic Reform Effect



(a) (b)

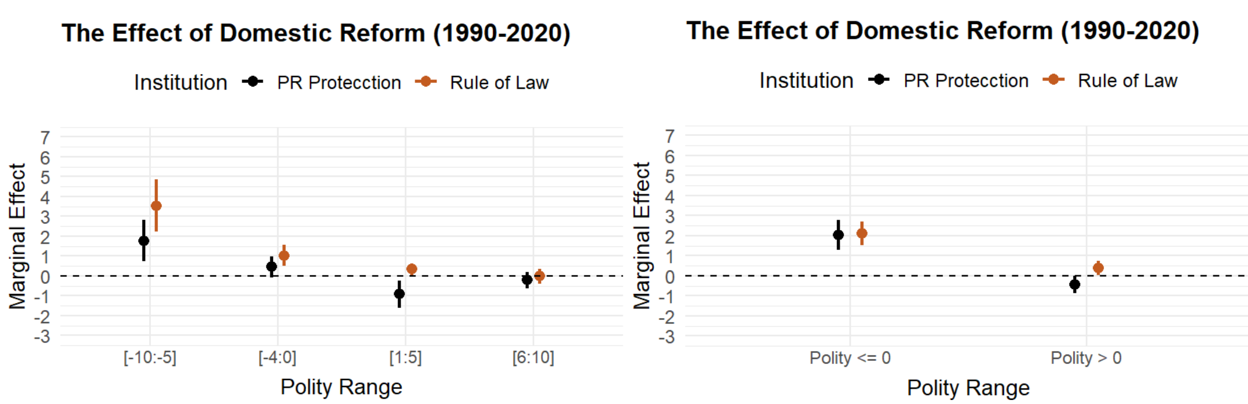
Figure C.13: The Effects of Domestic Reform by WTO-related Category. *Note*: (a) plots the effects of within-dyad changes of institutions by WTO-related Category for democracies only (Polity > 0 in 2000). I only include developing countries (GDP per capital lower than \$20,000 in 2000) to focus on institutional reform. (b) plots the effects of within-dyad changes of institutions for autocracies only. "Joiner" means a country joined the WTO during 1990-2020.

D Mechanism Tests

E Robust Tests

E.1 The Domestic Reform Effect

I now test whether or not domestic reform favors autocracies. I also use gravity models to test the effect of domestic reform on trade. VDem's PR protection and rule of law are used to measure institutional levels. I exploit within-dyad variation with dyad and year fixed-effects which control for possible time- and dyad-invariant confounders. As reform is mostly for developing countries, I focus on those with GDP per capita below $20,000 in 2000, which results in 165 countries. For the case of post-1990 trade expansion, I also test the case of outsiders to align with my theory on post-1990 globalization: new-joiners (who joined the WTO after 1990) and never-joiners – 94 total with 30 autocracies. I assign countries into four ranges based on Polity score in 2000 as Polity is quite stable for post-1990 period. Institutions are lagged by one year.



(a) Insiders + Outsiders

(b) Outsiders

Figure E.14: The Effects of Domestic Reform by Polity. *Note*: (a) plots the effects of within-dyad institutional changes across ranges of Polity, for all developing countries (GDP per capital below $20,000 in 2000). (b) plots the same graph, but only for outsiders (new-joiners and never-joiners). Results are consistent for including both institutional measures.

As shown in Figure E.14a, the effects of domestic reform among developing countries during the period of 1990-2020 are generally higher for more autocratic states.[52] This pattern is more

[52]I don't control for WTO membership to avoid the post-treatment bias. Similar effects remain with WTO member-

apparent in Figure E.14b, where I remove those who already joined the WTO before 1990. The likely interpretation is that the influx of many well-performing autocracies into the global trade system may exert significant stress to more hands-off, open-market democracies, with an exemplary case being the China shock. The result is consistent with the hypothesis that autocratic advantages amplify the effect of domestic reform. Admittedly, autocracies may also have increased marginal returns due to lower starting institutional levels. However, developing democracies' institutions were not significantly higher: 0.6 vs. 0.45 (autocracy) for PR protection in 1990 (see Figure 7), which cannot explain why democracy's reform effect is near zero.

Trade Integration as the Scope Condition

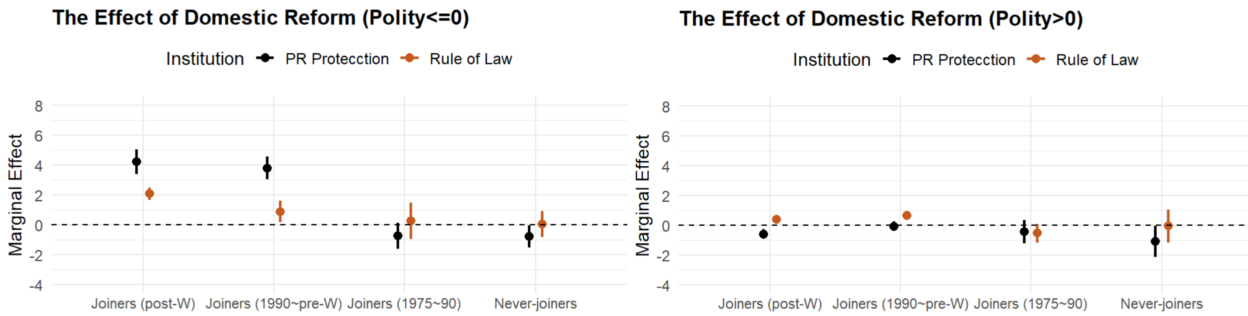
As the effect of domestic reform is only substantially positive for autocratic outsiders (Figure E.14b), I focus only on the dyad-years in which exporter is an autocracy in 2000.[53] Assuming the main gravity model that controls for trade's determinants is still valid, I fit a model including an interaction term to estimate effects for two strata: joiners (post-1990 pre-WTO period) and joiners (post-WTO period), for both autocracy and democracy, respectively:

$$Export_{ijt} = \beta Institution_{i,t-1} \times Polity_i \times WTO_{it} + \delta \mathbf{X}_{ijt} + \gamma_{ij} + \eta_t + \epsilon_{ijt}$$

where $\mathbf{X}_{ijt}$ denotes dyad-level covariates. $Institution_{i,t-1} \times WTO_{it} \times Polity_i$ captures the effect of institution (lagged) moderated by WTO period classified by Polity in 2000. $\mathbf{X}_{ijt}$ denotes dyad-level covariates. I include dyad and year fixed-effects. Additionally, I estimate the non-interaction version of the model for joiners' pre-1990 reform period (1975-1989) before they were substantively engaged, and never-joiners.

ship.

[53] I include three years earlier (i.e., 1987-89) to allow for pre-wto years for those joined in the early 1990s, though no inclusion doesn't affect results.



(a) Autocracies (Polity ≤ 0 in 2000)

(b) Democracies

Figure E.15: The Effects of Domestic Reform by Period. *Note*: Joiners are those joining the WTO after 1990. "W" denotes the WTO.

Figure E.15a displays the effects of domestic reform for four categories. For PR protection, every 0.1 increase leads to 65.3% in exports for joiners' pre-WTO period, 54.3% change for joiners' post-WTO periods, and almost no change for joiners' pre-1990 period (1975-1989) and never-joiners. For rule of law, the effects are 22.2%, 13.6%, null, and null, respectively. By comparison, democracies see almost null effects for all categories (Figure E.15b). Note that for joiners in the pre-WTO period (post-1990), as mentioned, many already enjoyed globalization benefits through semi-engagement, such as MFNs and PTAs, RTAs, spill-over from joiners, and future WTO prospect, so we expect positive effects compared to the non-engagement counterfactual.[54] Meanwhile, joiners' post-WTO period tend to have higher institutional level than pre-WTO period, which may have diminishing marginal returns of reforms, making the observed effect lower than that with a similar baseline as the pre-WTO period. In other words, WTO period may lead to higher effect than observed. Finally, for never-joiners, their effects are always null, although they may have other systemic differences. Overall, the results support my theory – autocratic advantages manifest conditional on the engagement by the global trade regime.

F Qualitative Tests

[54] I don't use lead WTO dummies as two periods still differ substantively.